

Shot point notebook

Robot · ABT CPSA 2026 · 25 clays

What you have in your hands

A complete round, clay by clay, exactly as the shooter would see it from the station. For each clay the whole trajectory is drawn, with five moments marked on it, spread across the window in which the clay is still within reach, plus one that is already late.

At each moment it tells you where the clay is and how much lead you need to break it, measured as the eye sees it. It is meant for planning the round before you go to the ground: what you shoot first, how much you lead, and what happens to you if you are late.

How to read each sheet

- The large figure is the lead as the eye perceives it, perpendicular to the line of sight. Below it also appears in clay widths and in degrees.
- The real three-dimensional separation is in small print: it includes the depth, which the eye does not see, and that is why it is a much larger number.
- Every sheet shares the same scale, so leads are comparable from one clay to another. The header says how much the view has been turned and with what framing.
- The colour of each moment warns you about distance: within reach, at the limit, or already beyond useful range.

What it is not

- It is not a field measurement: these are figures from a teaching model, with its simplifications. The last page explains this in detail.
- It does not certify an installation or a trajectory. In competition the authorised officials always decide.
- It does not replace an instructor, the rules of the ground, or the legal requirements.

Generated with these settings

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral

Every lead in this notebook is valid only for that combination. With another cartridge, another rib, another choke or with wind, the figures change and this notebook stops being useful: you have to generate a new one.

You do that from the simulator, changing whatever you want and pressing «Generate notebook» again. On each sheet, the settings that are not the start-up ones appear highlighted in colour.

The project

Plato is a free simulator and book for learning clay target shooting, written by Roberto Canales Mora. The book explains the ground, safety, the disciplines and the geometry of lead; the simulator lets you repeat the same clay changing one single variable.

Project, contact and repository on GitHub:

www.robertocanales.com/proyectos#plato
rcanalescoder.github.io/Plato

What is on the following pages

- An index of the round: a table with station, machine, angle, height, fall and lead for each clay, and a plan view with all the trajectories together.
- Una ficha por plato, 25 en total, en el orden en que los verías durante la tirada.
- The notice of use and the licence, at the end. If you print only part of it, print that sheet too.

Notice: the figures in this notebook are an estimate from a teaching model, not a field measurement. Read the notice of use and the licence on the last page before taking it to the ground.

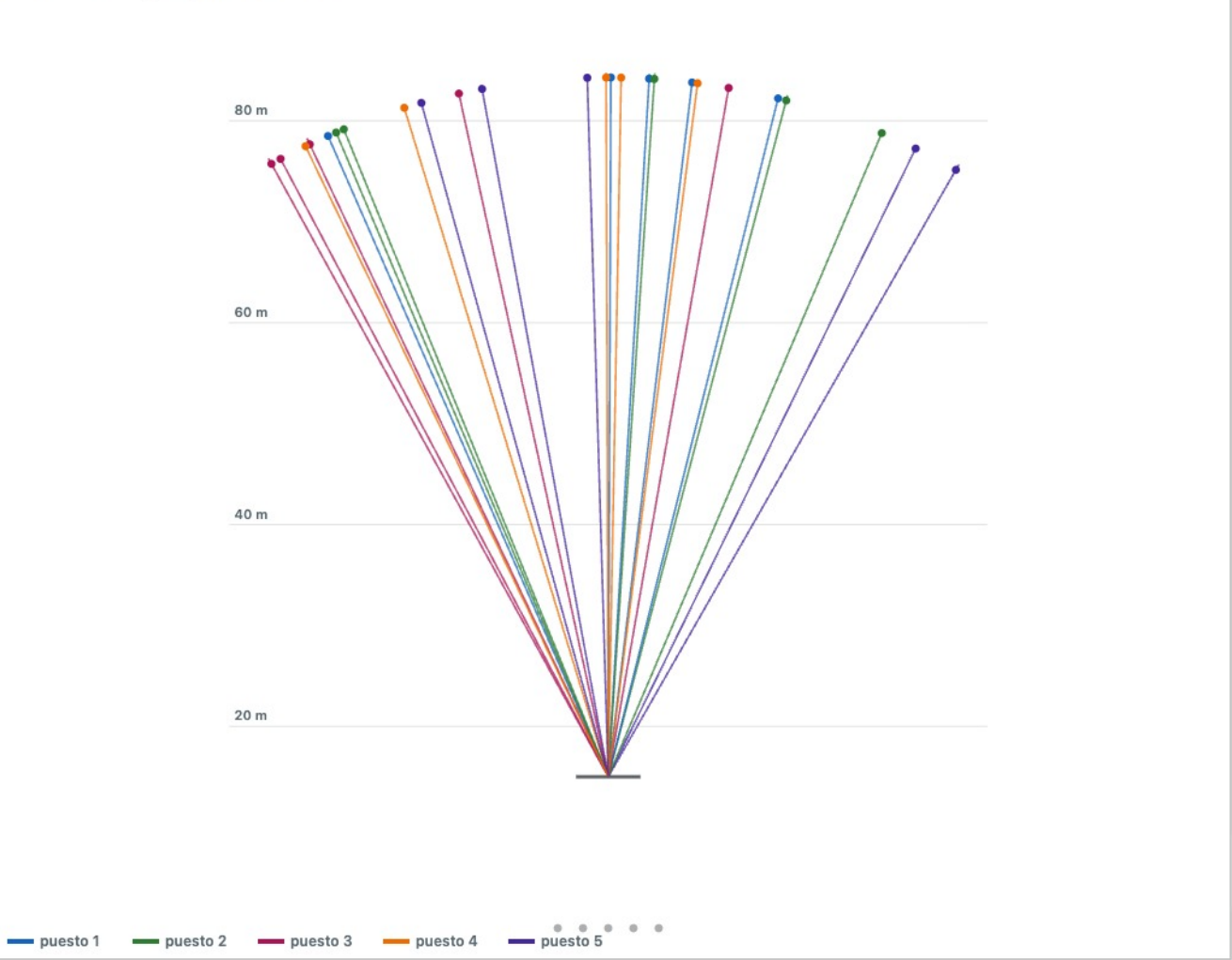
Shot point notebook

Robot · ABT CPSA 2026 · 25 clays

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral

#	station	mach.	angle	h. 10 m	fall	lead in the good window
1	1	1	+3°	2.1 m	70 m	52 cm · 4.7 platos · 0.92 s
2	2	1	-22°	3.1 m	70 m	78 cm · 7.1 platos · 1.17 s
3	3	1	-29°	2.4 m	70 m	82 cm · 7.5 platos · 1.04 s
4	4	1	-17°	3.4 m	70 m	18 cm · 1.7 platos · 1.17 s
5	5	1	-10°	2.2 m	70 m	27 cm · 2.5 platos · 0.94 s
6	1	1	+0°	2.3 m	70 m	59 cm · 5.4 platos · 0.96 s
7	2	1	+4°	2.4 m	70 m	18 cm · 1.6 platos · 0.99 s
8	3	1	-12°	1.8 m	70 m	45 cm · 4.1 platos · 0.87 s
9	4	1	-0°	2.8 m	70 m	26 cm · 2.4 platos · 1.06 s
10	5	1	+26°	2.2 m	70 m	132 cm · 12.0 platos · 1.03 s
11	1	1	-24°	1.5 m	70 m	164 cm · 14.9 platos · 0.84 s
12	2	1	+15°	3.4 m	70 m	13 cm · 1.2 platos · 1.18 s
13	3	1	-25°	2.4 m	70 m	74 cm · 6.8 platos · 1.02 s
14	4	1	+1°	3.3 m	70 m	26 cm · 2.4 platos · 1.16 s
15	5	1	-15°	3.3 m	70 m	8 cm · 0.7 platos · 1.13 s
16	1	1	+7°	1.8 m	70 m	45 cm · 4.1 platos · 0.85 s
17	2	1	-23°	3.5 m	70 m	75 cm · 6.8 platos · 1.24 s
18	3	1	-28°	2.0 m	70 m	90 cm · 8.1 platos · 0.95 s
19	4	1	-26°	2.4 m	70 m	49 cm · 4.4 platos · 1.01 s
20	5	1	+30°	2.9 m	70 m	120 cm · 10.9 platos · 1.18 s
21	1	1	+14°	3.4 m	70 m	11 cm · 1.0 platos · 1.15 s
22	2	1	+23°	3.3 m	70 m	34 cm · 3.1 platos · 1.16 s
23	3	1	+10°	2.5 m	70 m	29 cm · 2.6 platos · 1.02 s
24	4	1	+7°	3.0 m	70 m	44 cm · 4.0 platos · 1.11 s
25	5	1	-2°	1.5 m	70 m	71 cm · 6.4 platos · 0.79 s

Plan view · every clay in the scheme



The table uses the «good window» moment, the comfortable shot; each sheet details the other moments.

The lead is the one the shooter sees: the apparent separation between clay and aiming point, measured perpendicular to the line of sight.

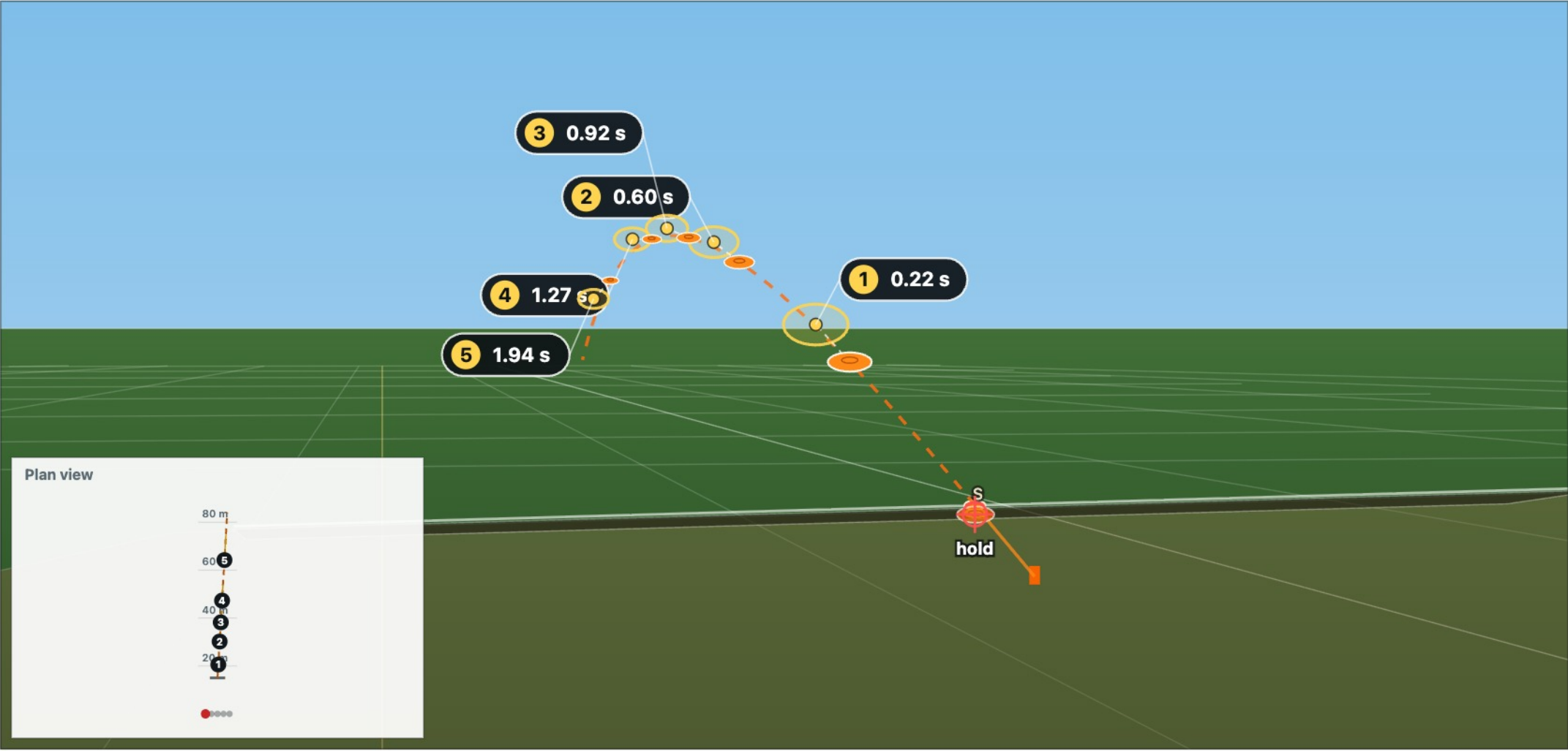
Red: the gun's hold point. Orange: the clay. Yellow: where to put the aiming point. Yellow ellipse: estimated pattern of the first shot.

Station 1 · Machine 1

angle +3° · height 2.1 m at 10 m · fall 70 m · 94 km/h · elevation 12.6°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -6° horizontally and +4° vertically



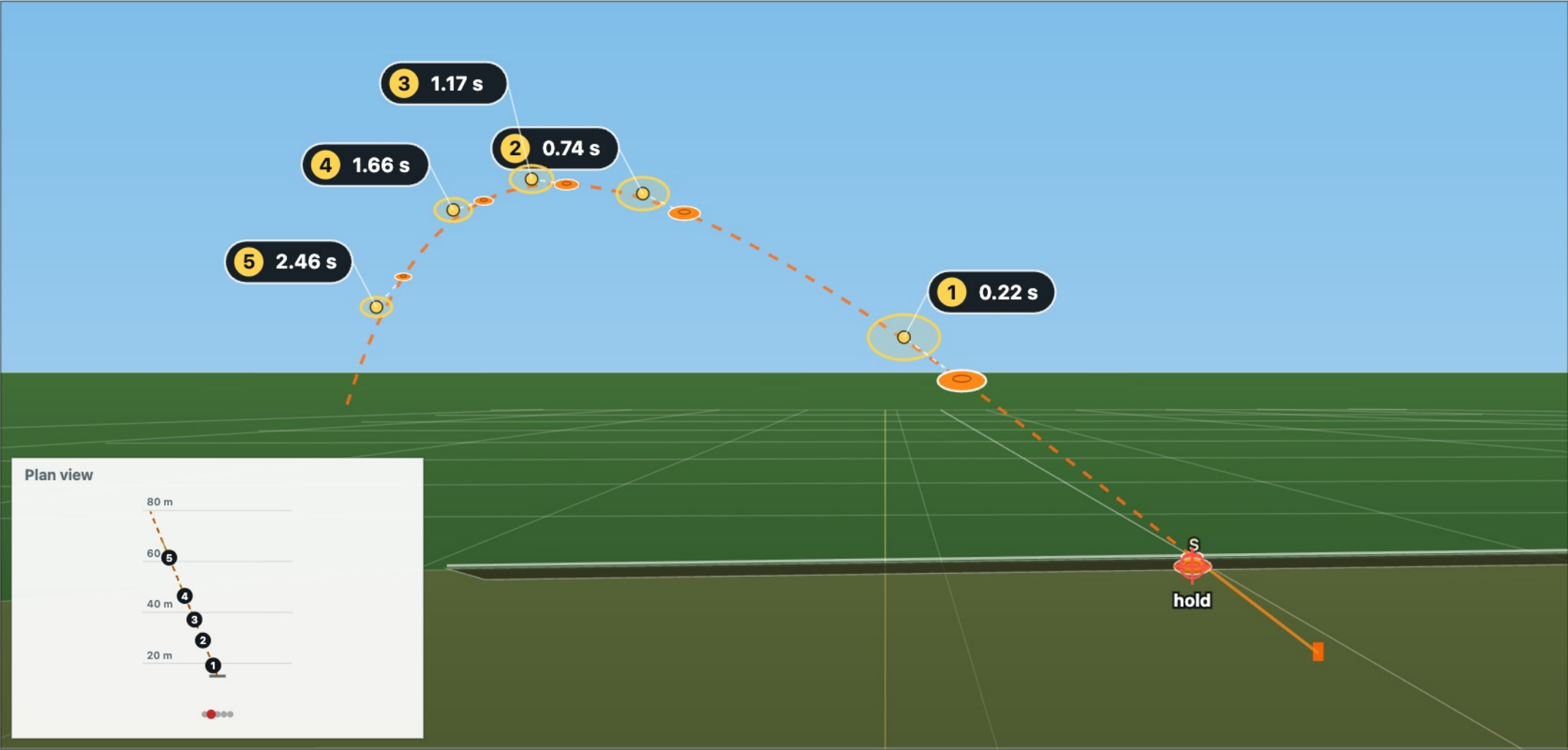
1 Release 0.22 s after release 63 cm ≈ 5.7 clays 44 cm left · 45 cm up -1.18° lateral · +1.21° vertical clay at 21 m · height 1.3 m · real separation 2.3 m first glimpse · within reach	2 Early 0.60 s after release 56 cm ≈ 5.1 clays 48 cm left · 28 cm up -0.89° lateral · +0.52° vertical clay at 31 m · height 2.8 m · real separation 3.6 m quick shot · within reach	3 Good window 0.92 s after release 52 cm ≈ 4.7 clays 52 cm left · 5 cm up -0.77° lateral · +0.07° vertical clay at 39 m · height 3.6 m · real separation 4.9 m the comfortable shot · at the limit	4 At the limit 1.27 s after release 67 cm ≈ 6.1 clays 57 cm left · 36 cm down -0.68° lateral · -0.43° vertical clay at 48 m · height 4.0 m · real separation 6.6 m last useful shot · at the limit	5 You are late 1.94 s after release 176 cm ≈ 16.0 clays 68 cm left · 162 cm down -0.60° lateral · -1.44° vertical clay at 65 m · height 3.4 m · real separation 10.7 m no longer reaching well · beyond useful range
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Station 2 · Machine 1

angle -22° · height 3.1 m at 10 m · fall 70 m · 78 km/h · elevation 19.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -13° horizontally and +6° vertically



Plan view



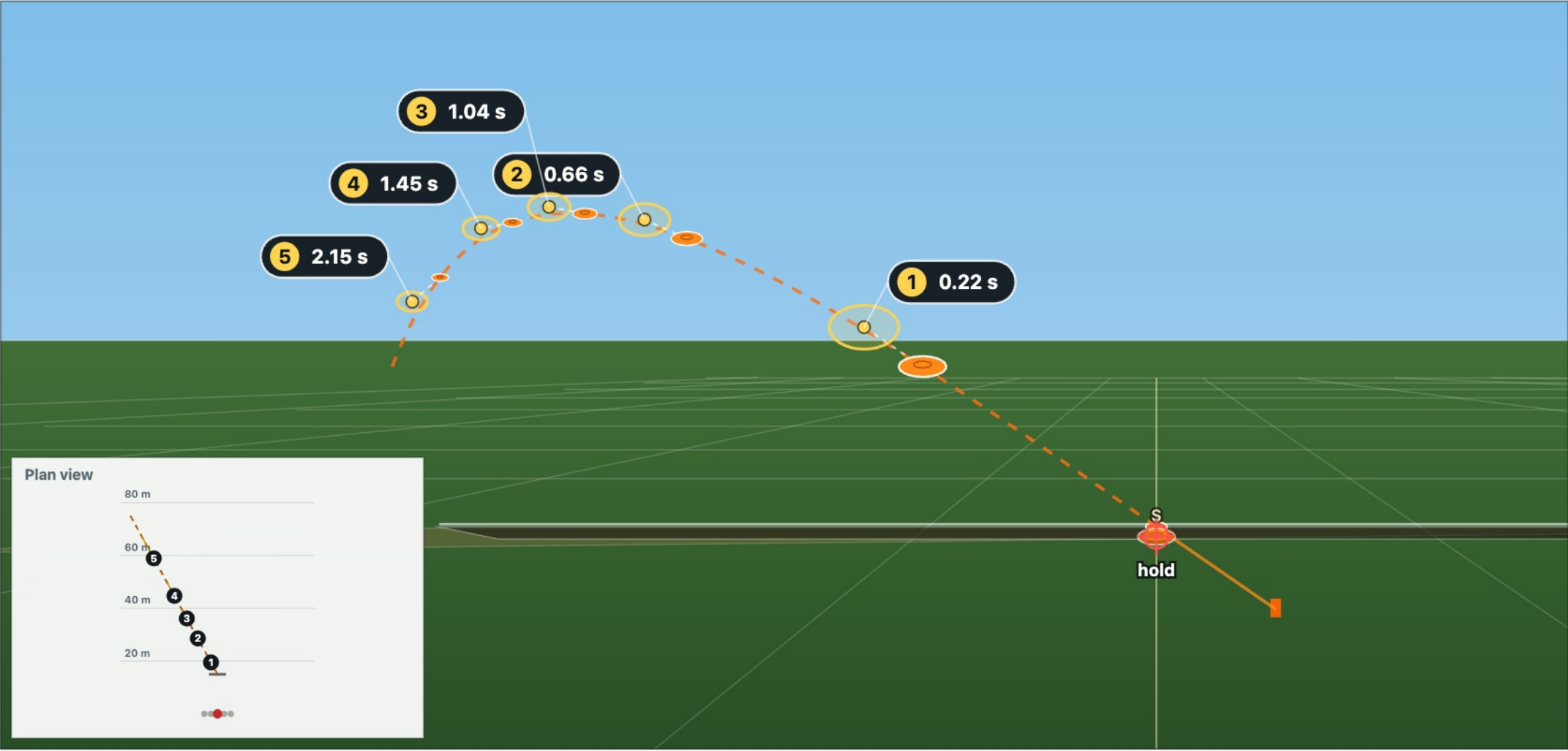
1 Release 0.22 s after release 80 cm ≈ 7.3 clays 65 cm left · 47 cm up -1.94° lateral · +1.39° vertical clay at 19 m · height 1.6 m · real separation 1.6 m first glimpse · within reach	2 Early 0.74 s after release 76 cm ≈ 6.9 clays 72 cm left · 26 cm up -1.41° lateral · +0.51° vertical clay at 29 m · height 4.3 m · real separation 2.7 m quick shot · within reach	3 Good window 1.17 s after release 78 cm ≈ 7.1 clays 78 cm left · 4 cm down -1.18° lateral · -0.06° vertical clay at 38 m · height 5.6 m · real separation 3.7 m the comfortable shot · within reach	4 At the limit 1.66 s after release 102 cm ≈ 9.3 clays 86 cm left · 56 cm down -1.03° lateral · -0.67° vertical clay at 48 m · height 6.2 m · real separation 5.1 m last useful shot · at the limit	5 You are late 2.46 s after release 221 cm ≈ 20.1 clays 101 cm left · 196 cm down -0.91° lateral · -1.76° vertical clay at 64 m · height 5.0 m · real separation 8.2 m no longer reaching well · beyond useful range
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Station 3 · Machine 1

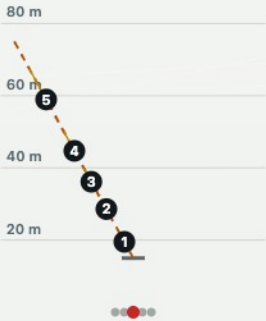
angle -29° · height 2.4 m at 10 m · fall 70 m · 87 km/h · elevation 14.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -12° horizontally and +5° vertically



Plan view



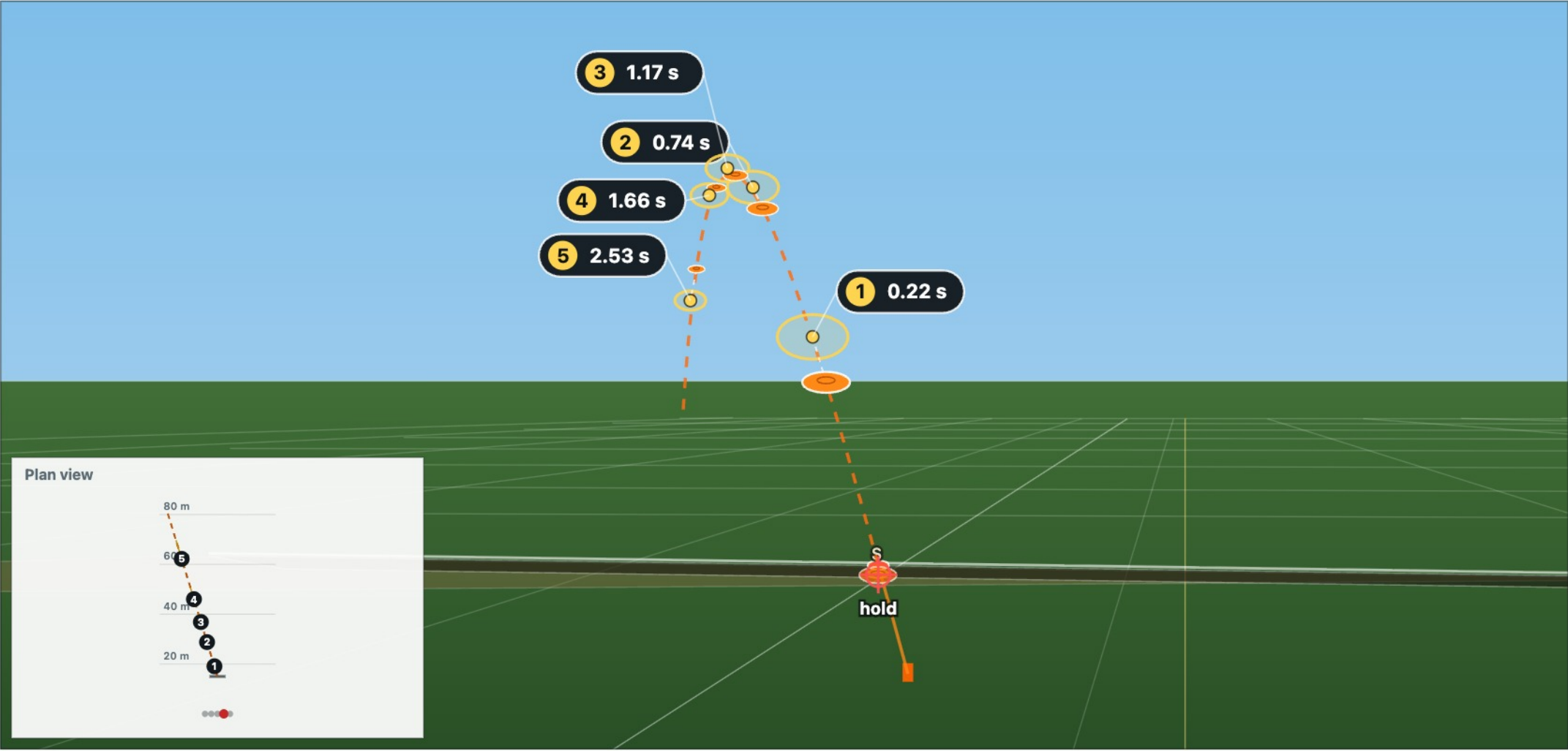
1 Release 0.22 s after release 81 cm ≈ 7.4 clays 68 cm left · 43 cm up -1.99° lateral · +1.26° vertical clay at 20 m · height 1.4 m · real separation 1.9 m first glimpse · within reach	2 Early 0.66 s after release 80 cm ≈ 7.2 clays 75 cm left · 25 cm up -1.46° lateral · +0.49° vertical clay at 30 m · height 3.4 m · real separation 3.1 m quick shot · within reach	3 Good window 1.04 s after release 82 cm ≈ 7.5 clays 82 cm left · 1 cm down -1.23° lateral · -0.02° vertical clay at 38 m · height 4.3 m · real separation 4.3 m the comfortable shot · at the limit	4 At the limit 1.45 s after release 102 cm ≈ 9.3 clays 90 cm left · 47 cm down -1.09° lateral · -0.56° vertical clay at 48 m · height 4.8 m · real separation 6.0 m last useful shot · at the limit	5 You are late 2.15 s after release 208 cm ≈ 18.9 clays 108 cm left · 177 cm down -0.97° lateral · -1.59° vertical clay at 64 m · height 3.9 m · real separation 9.6 m no longer reaching well · beyond useful range
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Station 4 · Machine 1

angle -17° · height 3.4 m at 10 m · fall 70 m · 75 km/h · elevation 20.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -3° horizontally and +6° vertically



Plan view



1 Release
0.22 s after release

52 cm

≈ 4.7 clays

15 cm left · 50 cm up

-0.46° lateral · +1.46° vertical

clay at 19 m · height 1.7 m · real separation 1.6 m

first glimpse · within reach

2 Early
0.74 s after release

34 cm

≈ 3.1 clays

17 cm left · 29 cm up

-0.33° lateral · +0.56° vertical

clay at 30 m · height 4.5 m · real separation 2.6 m

quick shot · within reach

3 Good window
1.17 s after release

18 cm

≈ 1.7 clays

18 cm left · 0 cm down

-0.28° lateral · -0.01° vertical

clay at 38 m · height 6.0 m · real separation 3.6 m

the comfortable shot · at the limit

4 At the limit
1.66 s after release

55 cm

≈ 5.0 clays

20 cm left · 51 cm down

-0.24° lateral · -0.61° vertical

clay at 48 m · height 6.8 m · real separation 4.9 m

last useful shot · at the limit

5 You are late
2.53 s after release

204 cm

≈ 18.6 clays

24 cm left · 203 cm down

-0.21° lateral · -1.80° vertical

clay at 65 m · height 5.7 m · real separation 8.0 m

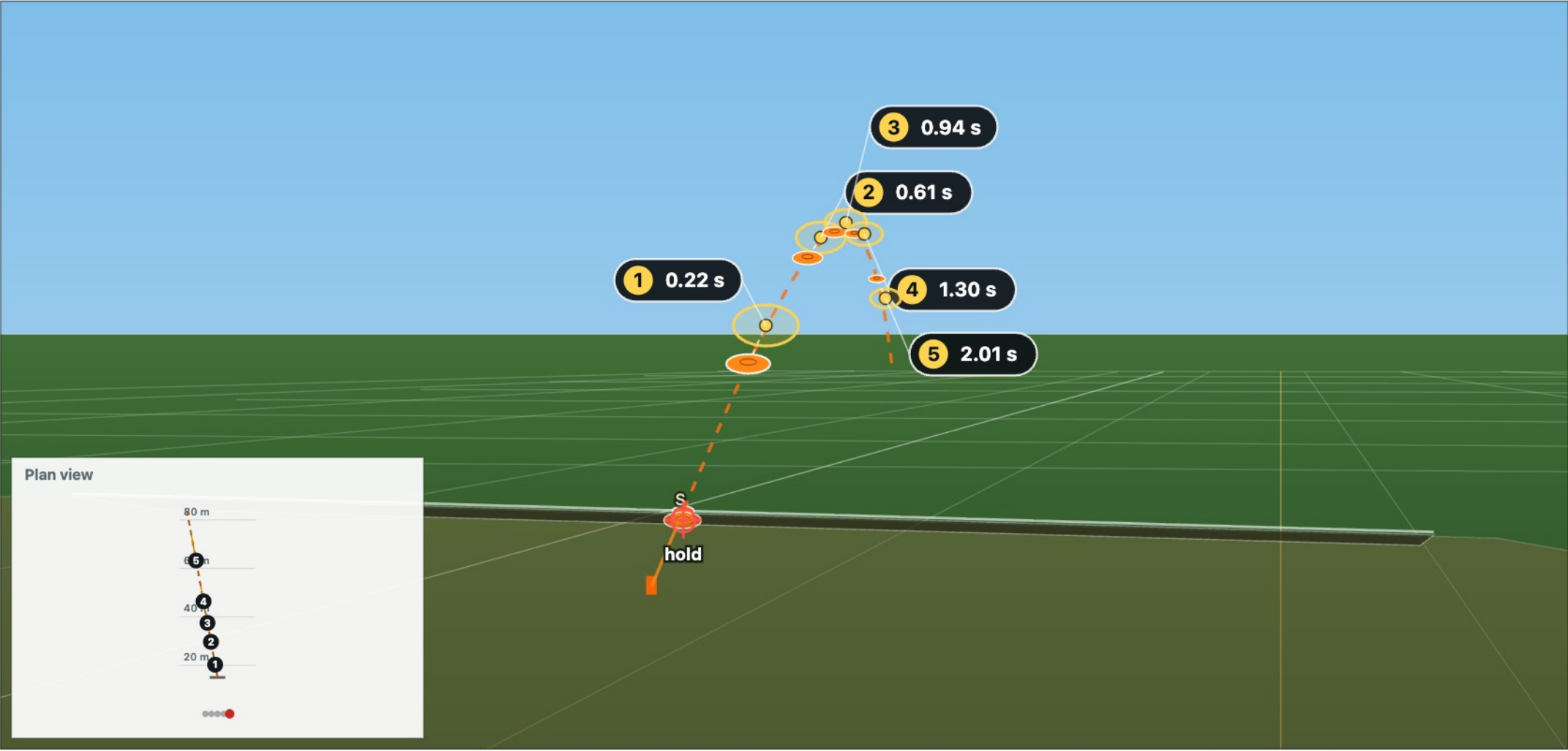
no longer reaching well · beyond useful range

Station 5 · Machine 1

angle -10° · height 2.2 m at 10 m · fall 70 m · 91 km/h · elevation 13.5°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +3° horizontally and +5° vertically



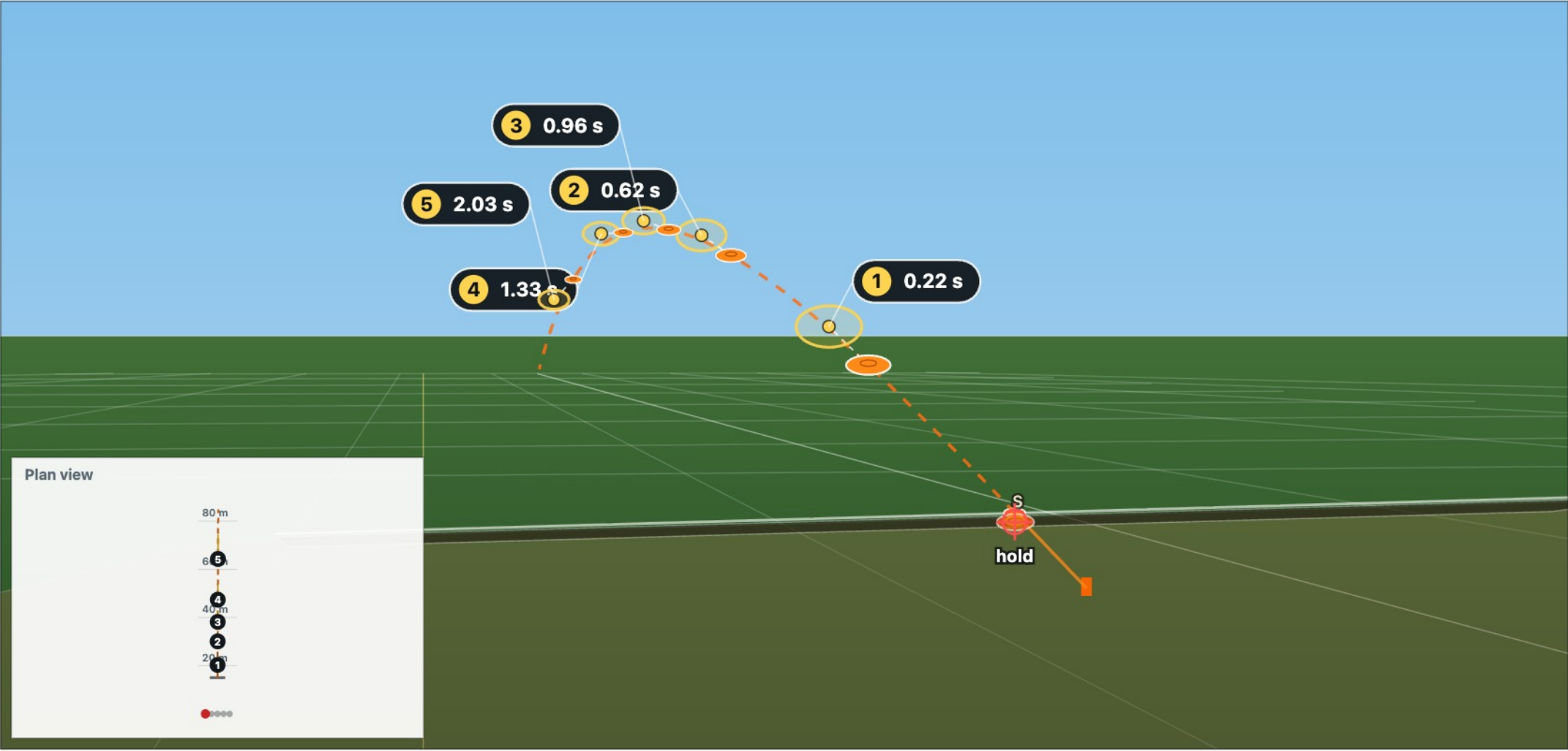
1 Release 0.22 s after release 51 cm ≈ 4.6 clays 23 cm right · 46 cm up +0.61° lateral · +1.24° vertical clay at 21 m · height 1.4 m · real separation 2.2 m first glimpse · within reach	2 Early 0.61 s after release 38 cm ≈ 3.5 clays 25 cm right · 29 cm up +0.46° lateral · +0.54° vertical clay at 31 m · height 3.0 m · real separation 3.5 m quick shot · within reach	3 Good window 0.94 s after release 27 cm ≈ 2.5 clays 27 cm right · 5 cm up +0.40° lateral · +0.07° vertical clay at 39 m · height 3.9 m · real separation 4.7 m the comfortable shot · at the limit	4 At the limit 1.30 s after release 46 cm ≈ 4.2 clays 29 cm right · 36 cm down +0.35° lateral · -0.43° vertical clay at 48 m · height 4.3 m · real separation 6.3 m last useful shot · at the limit	5 You are late 2.01 s after release 171 cm ≈ 15.6 clays 35 cm right · 168 cm down +0.31° lateral · -1.48° vertical clay at 65 m · height 3.7 m · real separation 10.4 m no longer reaching well · beyond useful range
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Station 1 · Machine 1

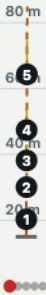
angle +0° · height 2.3 m at 10 m · fall 70 m · 90 km/h · elevation 13.7°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -7° horizontally and +5° vertically



Plan view



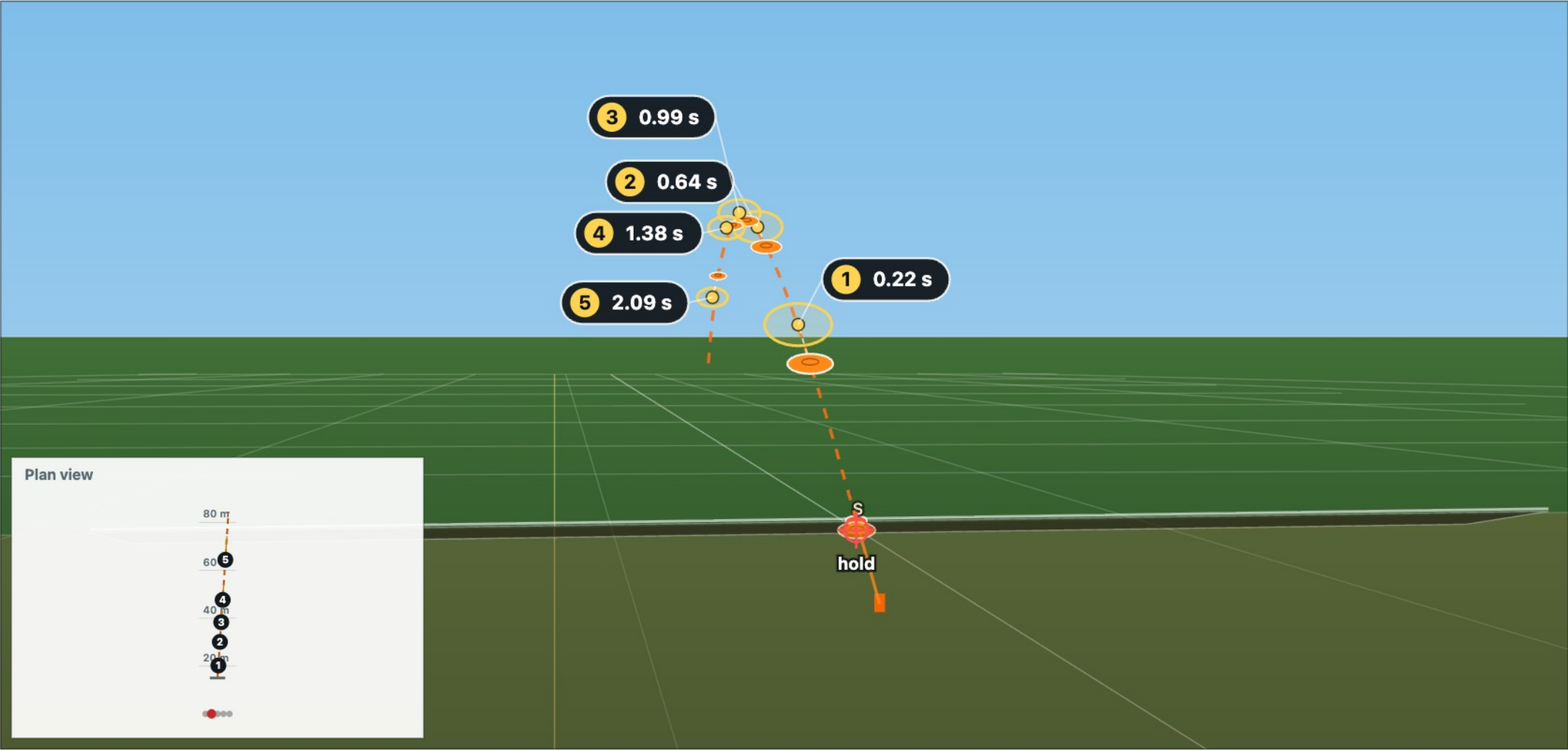
1 Release 0.22 s after release 67 cm ≈ 6.1 clays 50 cm left · 45 cm up -1.36° lateral · +1.24° vertical clay at 21 m · height 1.4 m · real separation 2.2 m first glimpse · within reach	2 Early 0.62 s after release 61 cm ≈ 5.6 clays 55 cm left · 28 cm up -1.03° lateral · +0.53° vertical clay at 31 m · height 3.0 m · real separation 3.4 m quick shot · within reach	3 Good window 0.96 s after release 59 cm ≈ 5.4 clays 59 cm left · 4 cm up -0.88° lateral · +0.06° vertical clay at 39 m · height 3.9 m · real separation 4.6 m the comfortable shot · at the limit	4 At the limit 1.33 s after release 75 cm ≈ 6.9 clays 65 cm left · 38 cm down -0.78° lateral · -0.46° vertical clay at 48 m · height 4.4 m · real separation 6.3 m last useful shot · at the limit	5 You are late 2.03 s after release 187 cm ≈ 17.0 clays 78 cm left · 170 cm down -0.69° lateral · -1.51° vertical clay at 65 m · height 3.7 m · real separation 10.3 m no longer reaching well · beyond useful range
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Station 2 · Machine 1

angle +4° · height 2.4 m at 10 m · fall 70 m · 88 km/h · elevation 14.4°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -2° horizontally and +5° vertically



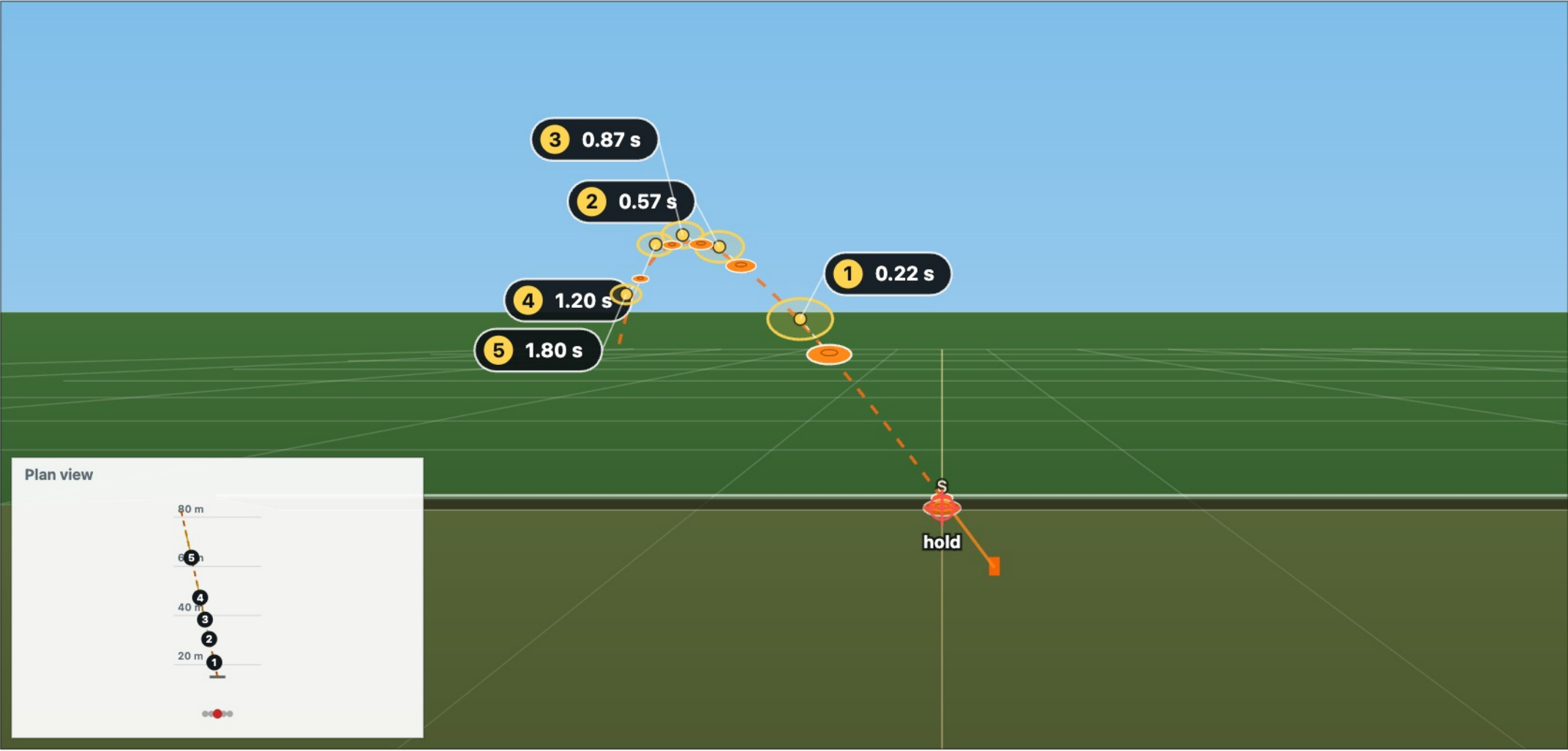
1 Release 0.22 s after release 47 cm ≈ 4.3 clays 15 cm left · 45 cm up -0.41° lateral · +1.26° vertical clay at 20 m · height 1.4 m · real separation 2.0 m first glimpse · within reach	2 Early 0.64 s after release 32 cm ≈ 2.9 clays 16 cm left · 27 cm up -0.31° lateral · +0.52° vertical clay at 30 m · height 3.2 m · real separation 3.3 m quick shot · within reach	3 Good window 0.99 s after release 18 cm ≈ 1.6 clays 18 cm left · 2 cm up -0.26° lateral · +0.03° vertical clay at 39 m · height 4.1 m · real separation 4.5 m the comfortable shot · at the limit	4 At the limit 1.38 s after release 45 cm ≈ 4.1 clays 19 cm left · 41 cm down -0.23° lateral · -0.49° vertical clay at 48 m · height 4.6 m · real separation 6.1 m last useful shot · at the limit	5 You are late 2.09 s after release 177 cm ≈ 16.1 clays 23 cm left · 176 cm down -0.21° lateral · -1.56° vertical clay at 65 m · height 3.9 m · real separation 10.1 m no longer reaching well · beyond useful range
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Station 3 · Machine 1

angle -12° · height 1.8 m at 10 m · fall 70 m · 101 km/h · elevation 10.7°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -5° horizontally and +4° vertically



Plan view



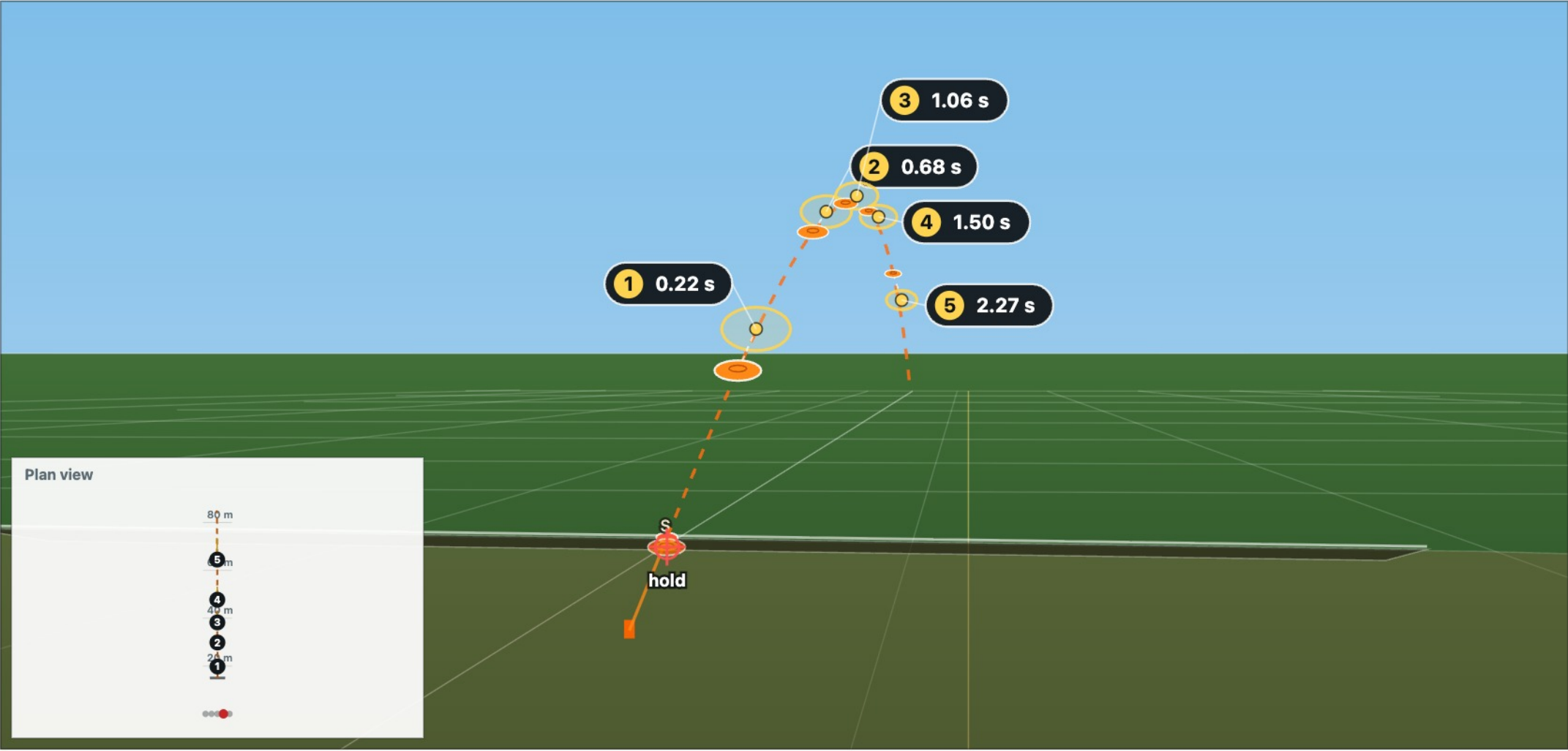
1 Release 0.22 s after release 56 cm ≈ 5.1 clays 37 cm left · 42 cm up -1.02° lateral · +1.15° vertical clay at 21 m · height 1.2 m · real separation 2.5 m first glimpse · within reach	2 Early 0.57 s after release 49 cm ≈ 4.4 clays 41 cm left · 27 cm up -0.77° lateral · +0.50° vertical clay at 31 m · height 2.5 m · real separation 3.9 m quick shot · within reach	3 Good window 0.87 s after release 45 cm ≈ 4.1 clays 45 cm left · 4 cm up -0.66° lateral · +0.06° vertical clay at 39 m · height 3.1 m · real separation 5.4 m the comfortable shot · at the limit	4 At the limit 1.20 s after release 60 cm ≈ 5.5 clays 49 cm left · 34 cm down -0.59° lateral · -0.41° vertical clay at 48 m · height 3.5 m · real separation 7.4 m last useful shot · at the limit	5 You are late 1.80 s after release 161 cm ≈ 14.6 clays 58 cm left · 150 cm down -0.51° lateral · -1.34° vertical clay at 64 m · height 2.9 m · real separation 11.6 m no longer reaching well · beyond useful range
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Station 4 · Machine 1

angle -0° · height 2.8 m at 10 m · fall 70 m · 82 km/h · elevation 16.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +4° horizontally and +5° vertically



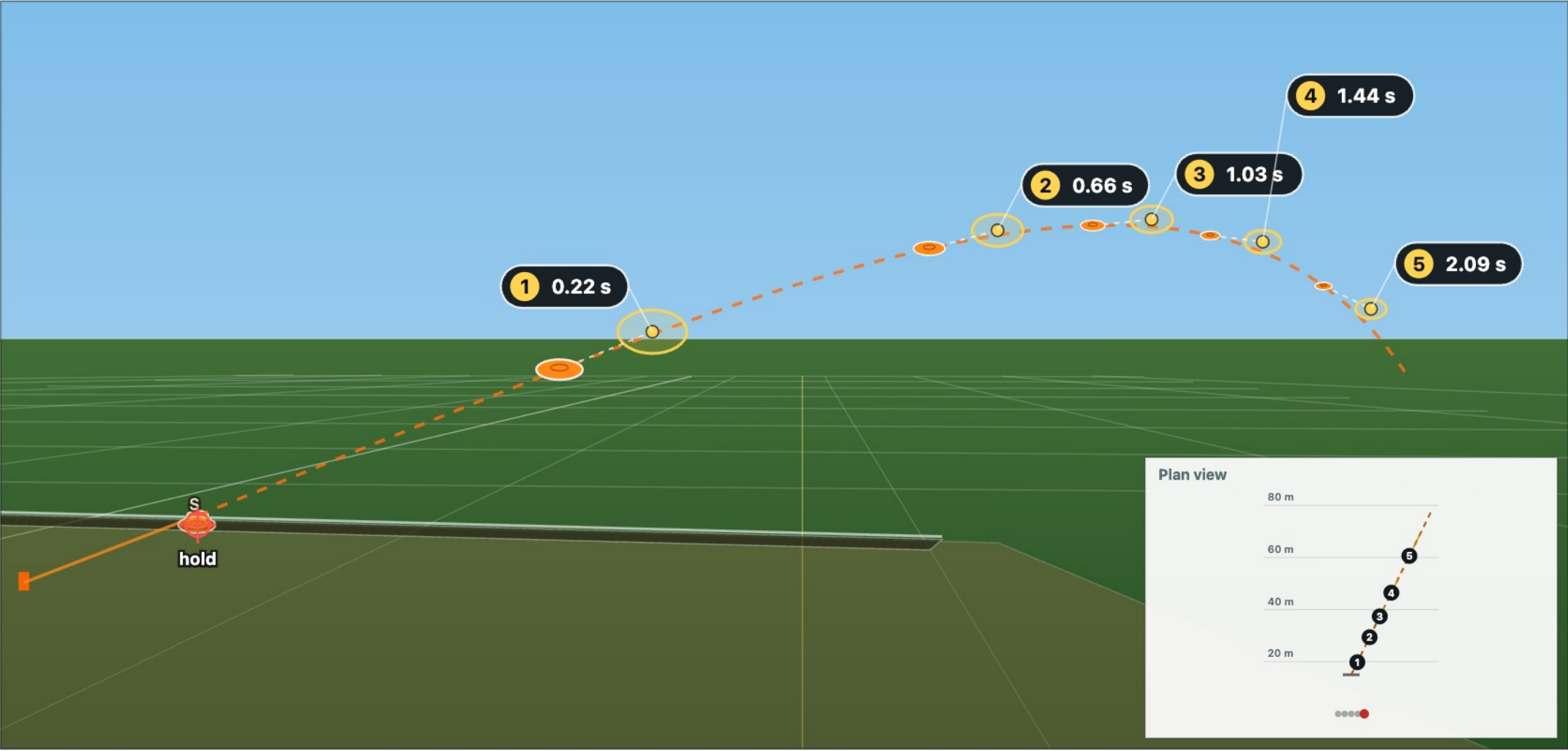
1 Release 0.22 s after release 51 cm ≈ 4.7 clays 22 cm right · 47 cm up +0.63° lateral · +1.34° vertical clay at 20 m · height 1.5 m · real separation 1.8 m first glimpse · within reach	2 Early 0.68 s after release 37 cm ≈ 3.3 clays 24 cm right · 28 cm up +0.46° lateral · +0.53° vertical clay at 30 m · height 3.7 m · real separation 3.0 m quick shot · within reach	3 Good window 1.06 s after release 26 cm ≈ 2.4 clays 26 cm right · 1 cm up +0.39° lateral · +0.01° vertical clay at 38 m · height 4.9 m · real separation 4.1 m the comfortable shot · at the limit	4 At the limit 1.50 s after release 54 cm ≈ 4.9 clays 29 cm right · 46 cm down +0.34° lateral · -0.55° vertical clay at 48 m · height 5.4 m · real separation 5.6 m last useful shot · at the limit	5 You are late 2.27 s after release 191 cm ≈ 17.3 clays 34 cm right · 188 cm down +0.30° lateral · -1.66° vertical clay at 65 m · height 4.5 m · real separation 9.1 m no longer reaching well · beyond useful range
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Station 5 · Machine 1

angle +26° · height 2.2 m at 10 m · fall 70 m · 90 km/h · elevation 13.7°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +18° horizontally and +5° vertically



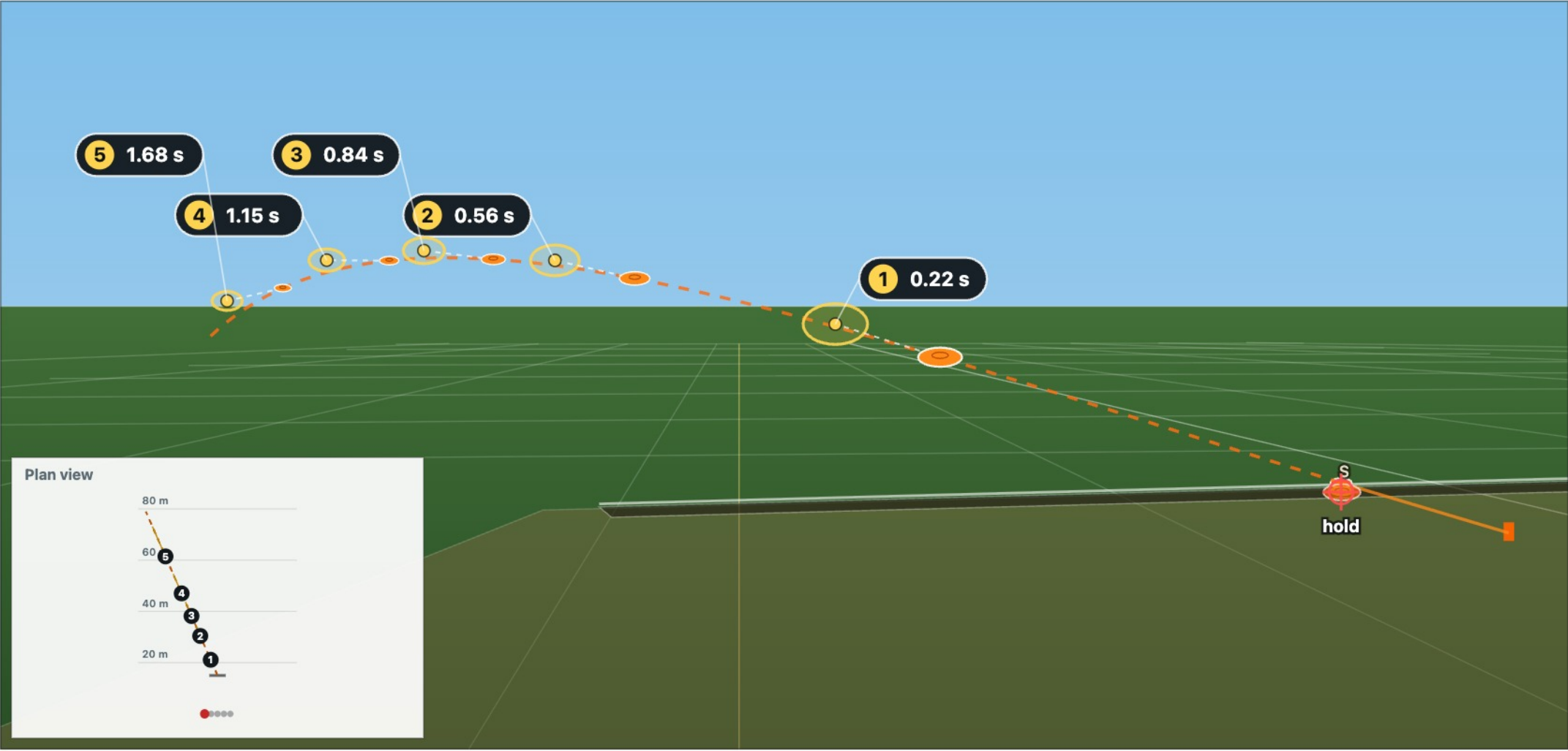
1 Release 0.22 s after release 117 cm ≈ 10.7 clays 109 cm right · 42 cm up +3.13° lateral · +1.20° vertical clay at 20 m · height 1.4 m · real separation 2.0 m first glimpse · within reach	2 Early 0.66 s after release 123 cm ≈ 11.2 clays 121 cm right · 24 cm up +2.34° lateral · +0.46° vertical clay at 30 m · height 3.2 m · real separation 3.2 m quick shot · within reach	3 Good window 1.03 s after release 132 cm ≈ 12.0 clays 132 cm right · 3 cm down +1.99° lateral · −0.04° vertical clay at 38 m · height 4.1 m · real separation 4.5 m the comfortable shot · within reach	4 At the limit 1.44 s after release 154 cm ≈ 14.0 clays 146 cm right · 49 cm down +1.75° lateral · −0.58° vertical clay at 48 m · height 4.4 m · real separation 6.3 m last useful shot · at the limit	5 You are late 2.09 s after release 243 cm ≈ 22.1 clays 173 cm right · 171 cm down +1.57° lateral · −1.55° vertical clay at 63 m · height 3.5 m · real separation 10.0 m no longer reaching well · beyond useful range
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Station 1 · Machine 1

angle -24° · height 1.5 m at 10 m · fall 70 m · 110 km/h · elevation 8.9°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -17° horizontally and +4° vertically



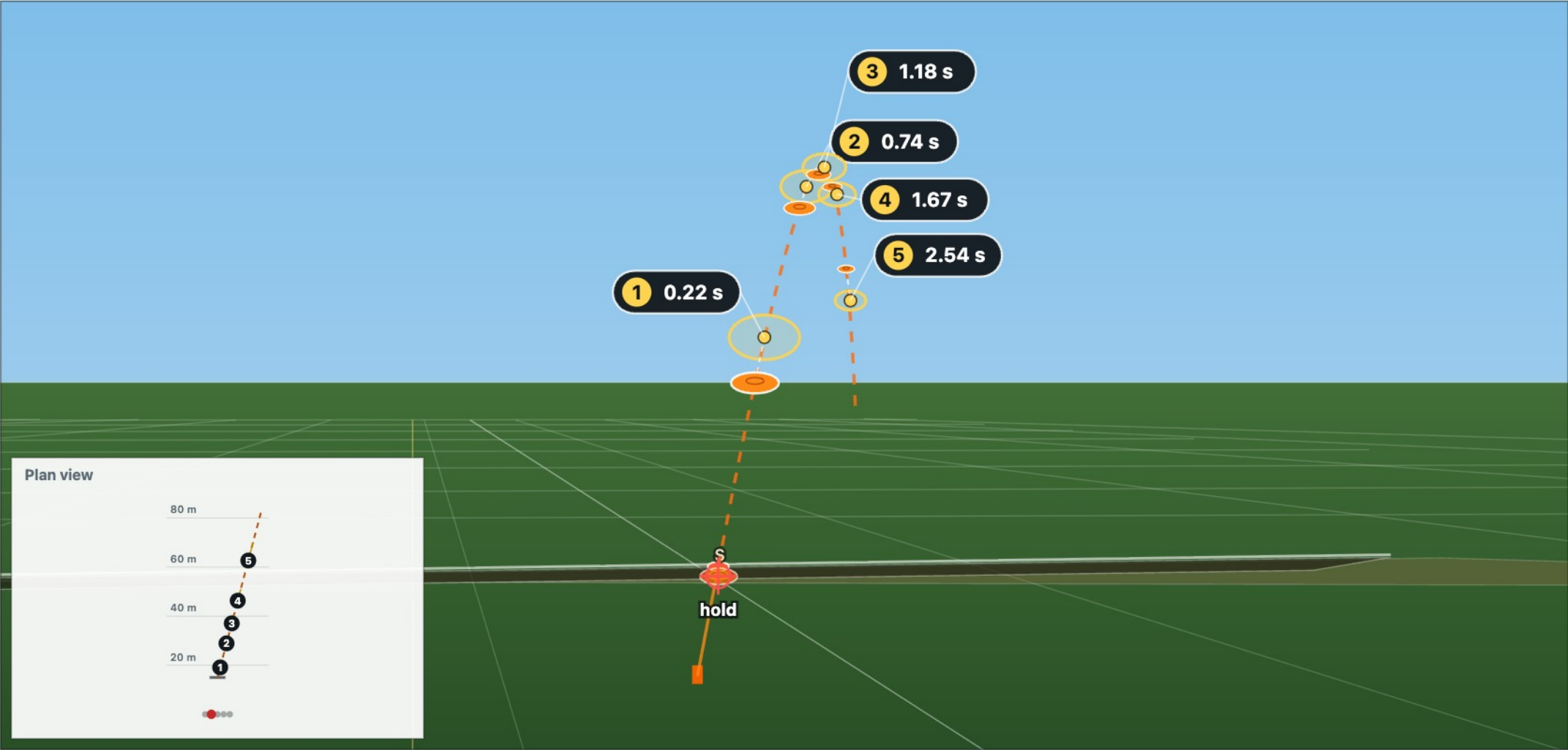
1 Release 0.22 s after release 141 cm ≈ 12.8 clays 135 cm left · 39 cm up -3.65° lateral · +1.06° vertical clay at 21 m · height 1.1 m · real separation 2.7 m first glimpse · within reach	2 Early 0.56 s after release 152 cm ≈ 13.8 clays 150 cm left · 25 cm up -2.81° lateral · +0.46° vertical clay at 30 m · height 2.2 m · real separation 4.3 m quick shot · within reach	3 Good window 0.84 s after release 164 cm ≈ 14.9 clays 164 cm left · 3 cm up -2.43° lateral · +0.04° vertical clay at 39 m · height 2.7 m · real separation 6.0 m the comfortable shot · at the limit	4 At the limit 1.15 s after release 186 cm ≈ 16.9 clays 182 cm left · 36 cm down -2.18° lateral · -0.43° vertical clay at 48 m · height 2.9 m · real separation 8.2 m last useful shot · at the limit	5 You are late 1.68 s after release 254 cm ≈ 23.1 clays 212 cm left · 139 cm down -1.92° lateral · -1.26° vertical clay at 63 m · height 2.3 m · real separation 12.8 m no longer reaching well · beyond useful range
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Station 2 · Machine 1

angle +15° · height 3.4 m at 10 m · fall 70 m · 75 km/h · elevation 20.9°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +2° horizontally and +6° vertically



Plan view



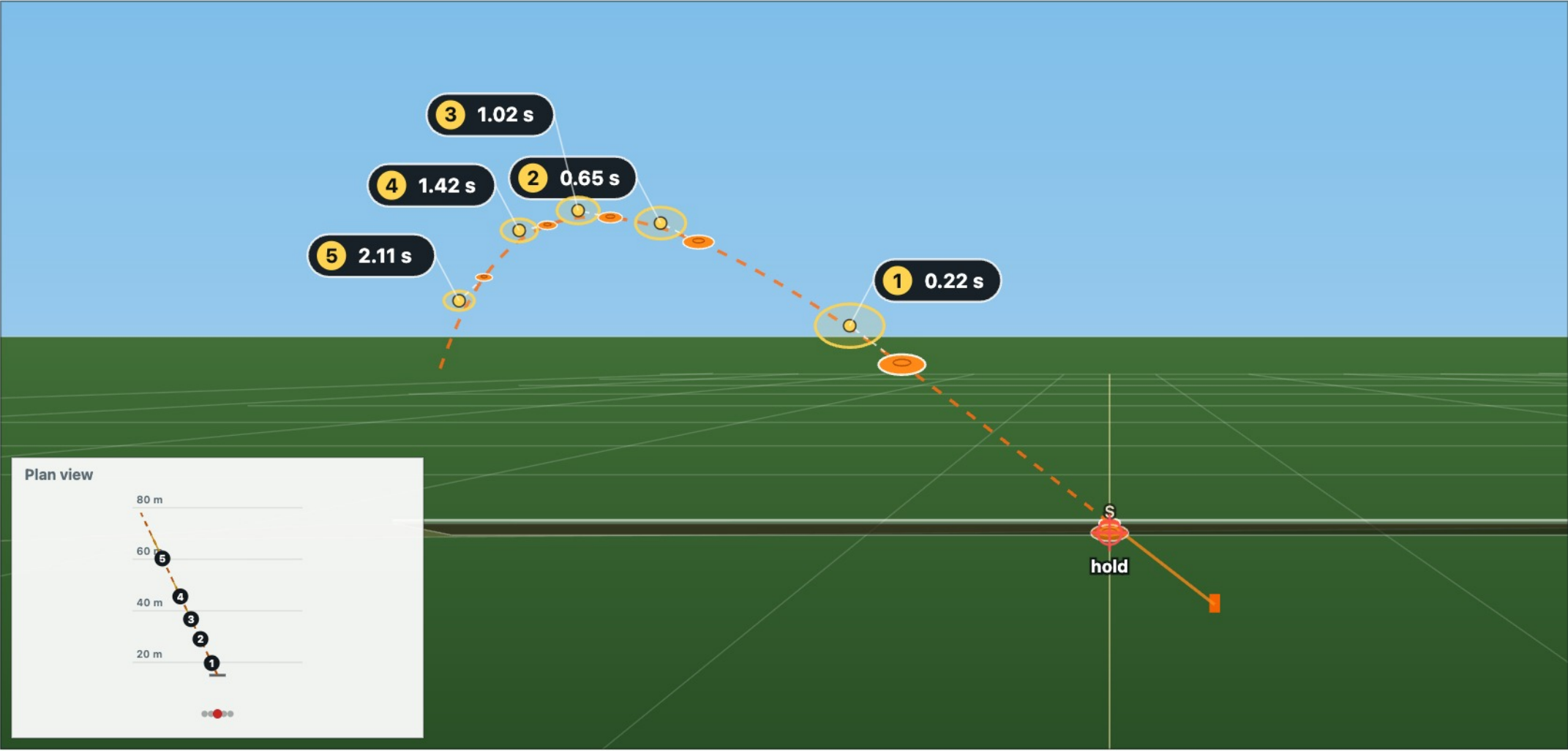
1 Release 0.22 s after release 51 cm ≈ 4.6 clays 11 cm right · 50 cm up +0.32° lateral · +1.46° vertical clay at 19 m · height 1.7 m · real separation 1.6 m first glimpse · within reach	2 Early 0.74 s after release 32 cm ≈ 2.9 clays 12 cm right · 29 cm up +0.23° lateral · +0.56° vertical clay at 30 m · height 4.5 m · real separation 2.6 m quick shot · within reach	3 Good window 1.18 s after release 13 cm ≈ 1.2 clays 13 cm right · 0 cm down +0.19° lateral · -0.00° vertical clay at 38 m · height 6.0 m · real separation 3.6 m the comfortable shot · at the limit	4 At the limit 1.67 s after release 53 cm ≈ 4.8 clays 14 cm right · 51 cm down +0.17° lateral · -0.61° vertical clay at 48 m · height 6.8 m · real separation 4.8 m last useful shot · at the limit	5 You are late 2.54 s after release 204 cm ≈ 18.6 clays 17 cm right · 204 cm down +0.15° lateral · -1.81° vertical clay at 65 m · height 5.7 m · real separation 7.9 m no longer reaching well · beyond useful range
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Station 3 • Machine 1

angle -25° • height 2.4 m at 10 m • fall 70 m • 88 km/h • elevation 14.4°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s • rib 0% (0.00°) • wind 0.0 m/s • delay 0.03 s • pattern 1 3/4 • hold neutral
framing 2.16x over the base field of 48° vertical • view turned -10° horizontally and +5° vertically



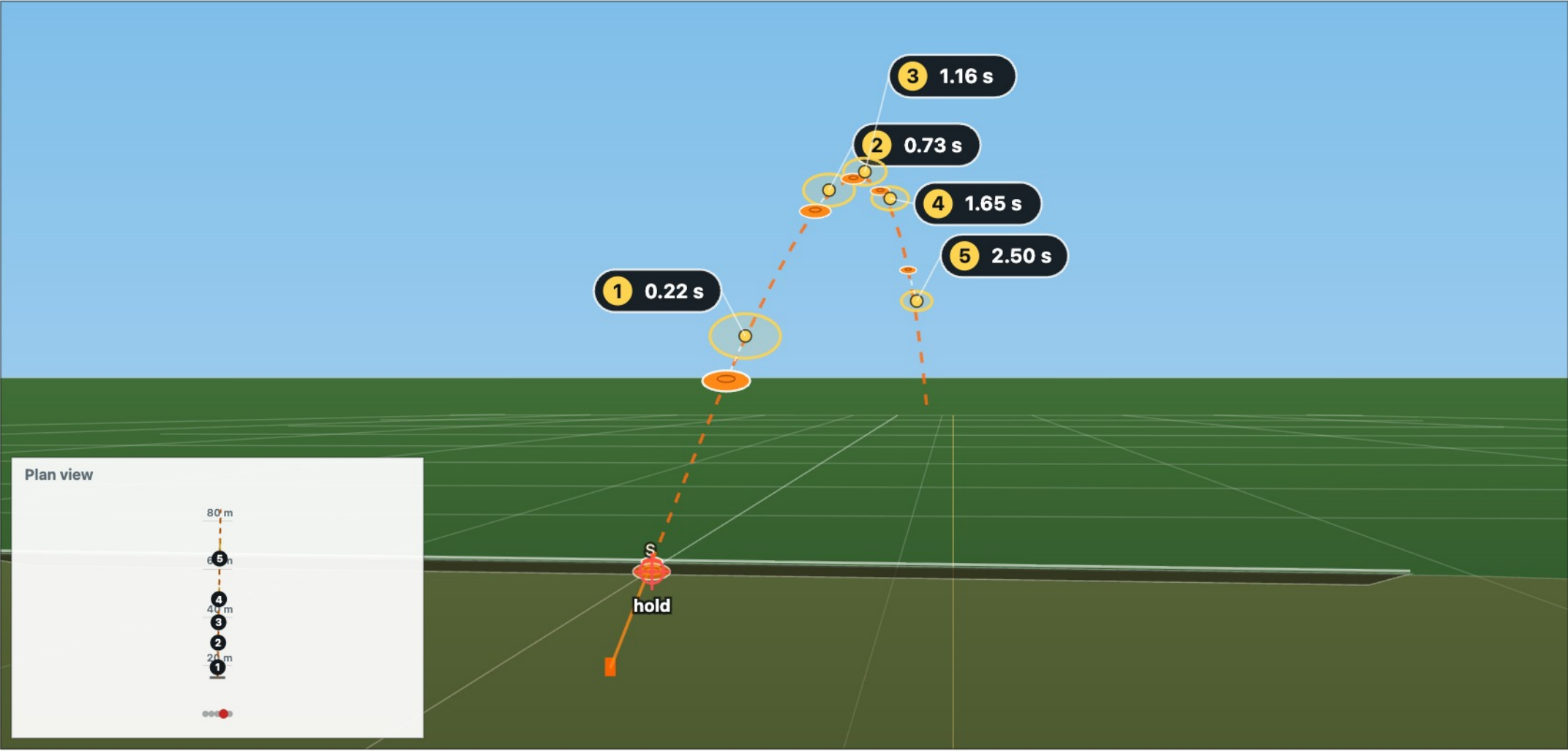
1 Release 0.22 s after release 76 cm ≈ 6.9 clays 62 cm left • 43 cm up -1.79° lateral • +1.25° vertical clay at 20 m • height 1.4 m • real separation 2.0 m first glimpse • within reach	2 Early 0.65 s after release 73 cm ≈ 6.6 clays 68 cm left • 25 cm up -1.31° lateral • +0.49° vertical clay at 30 m • height 3.2 m • real separation 3.2 m quick shot • within reach	3 Good window 1.02 s after release 74 cm ≈ 6.8 clays 74 cm left • 1 cm down -1.11° lateral • -0.01° vertical clay at 38 m • height 4.2 m • real separation 4.4 m the comfortable shot • at the limit	4 At the limit 1.42 s after release 94 cm ≈ 8.5 clays 82 cm left • 46 cm down -0.98° lateral • -0.55° vertical clay at 48 m • height 4.6 m • real separation 6.1 m last useful shot • at the limit	5 You are late 2.11 s after release 202 cm ≈ 18.3 clays 98 cm left • 176 cm down -0.88° lateral • -1.58° vertical clay at 64 m • height 3.8 m • real separation 9.9 m no longer reaching well • beyond useful range
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Station 4 · Machine 1

angle +1° · height 3.3 m at 10 m · fall 70 m · 76 km/h · elevation 20.3°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +4° horizontally and +6° vertically



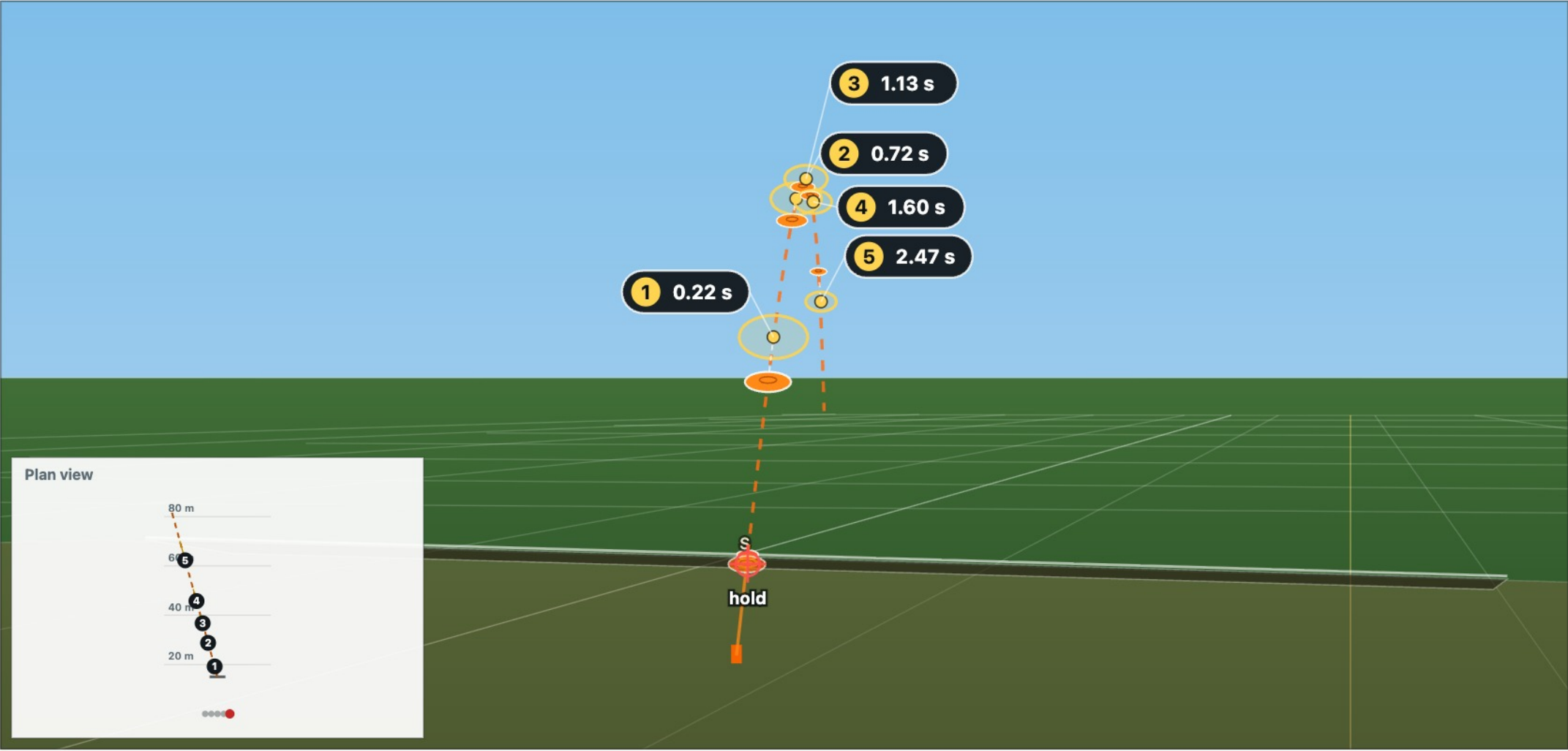
1 Release 0.22 s after release 54 cm ≈ 4.9 clays 22 cm right · 49 cm up +0.65° lateral · +1.44° vertical clay at 19 m · height 1.7 m · real separation 1.6 m first glimpse · within reach	2 Early 0.73 s after release 38 cm ≈ 3.4 clays 24 cm right · 29 cm up +0.47° lateral · +0.55° vertical clay at 30 m · height 4.4 m · real separation 2.6 m quick shot · within reach	3 Good window 1.16 s after release 26 cm ≈ 2.4 clays 26 cm right · 1 cm down +0.39° lateral · -0.01° vertical clay at 38 m · height 5.9 m · real separation 3.6 m the comfortable shot · at the limit	4 At the limit 1.65 s after release 58 cm ≈ 5.3 clays 29 cm right · 51 cm down +0.35° lateral · -0.61° vertical clay at 48 m · height 6.6 m · real separation 4.9 m last useful shot · at the limit	5 You are late 2.50 s after release 204 cm ≈ 18.6 clays 34 cm right · 201 cm down +0.30° lateral · -1.79° vertical clay at 65 m · height 5.5 m · real separation 8.1 m no longer reaching well · beyond useful range
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Station 5 · Machine 1

angle -15° · height 3.3 m at 10 m · fall 70 m · 76 km/h · elevation 20.0°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +1° horizontally and +6° vertically



Plan view



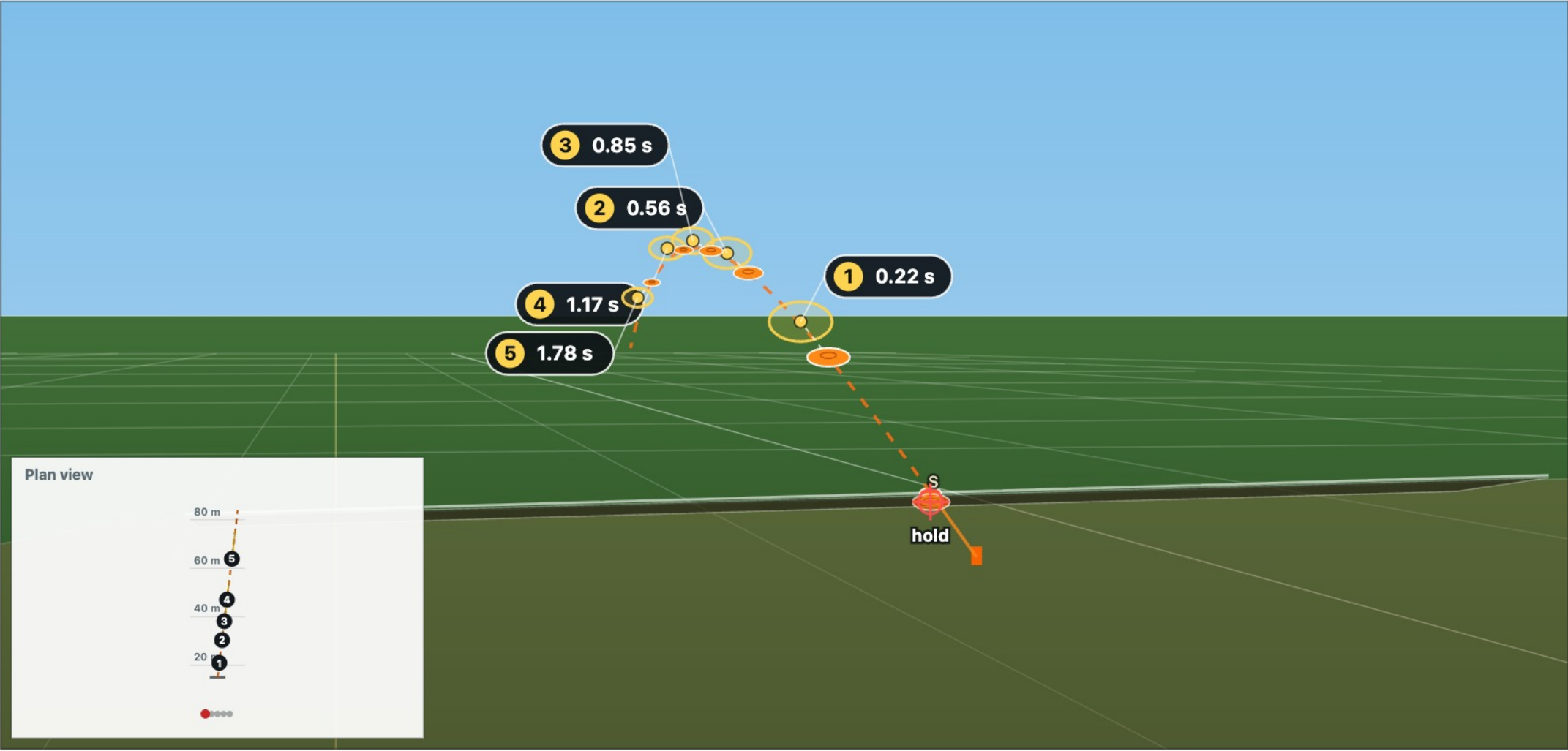
1 Release 0.22 s after release 51 cm ≈ 4.7 clays 7 cm right · 51 cm up +0.19° lateral · +1.44° vertical clay at 20 m · height 1.7 m · real separation 1.7 m first glimpse · within reach	2 Early 0.72 s after release 31 cm ≈ 2.9 clays 7 cm right · 31 cm up +0.14° lateral · +0.58° vertical clay at 30 m · height 4.3 m · real separation 2.7 m quick shot · within reach	3 Good window 1.13 s after release 8 cm ≈ 0.7 clays 8 cm right · 2 cm up +0.12° lateral · +0.03° vertical clay at 39 m · height 5.7 m · real separation 3.7 m the comfortable shot · at the limit	4 At the limit 1.60 s after release 48 cm ≈ 4.3 clays 9 cm right · 47 cm down +0.10° lateral · -0.56° vertical clay at 48 m · height 6.5 m · real separation 5.0 m last useful shot · at the limit	5 You are late 2.47 s after release 200 cm ≈ 18.2 clays 10 cm right · 200 cm down +0.09° lateral · -1.77° vertical clay at 65 m · height 5.5 m · real separation 8.2 m no longer reaching well · beyond useful range
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Station 1 · Machine 1

angle +7° · height 1.8 m at 10 m · fall 70 m · 101 km/h · elevation 10.7°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -5° horizontally and +4° vertically



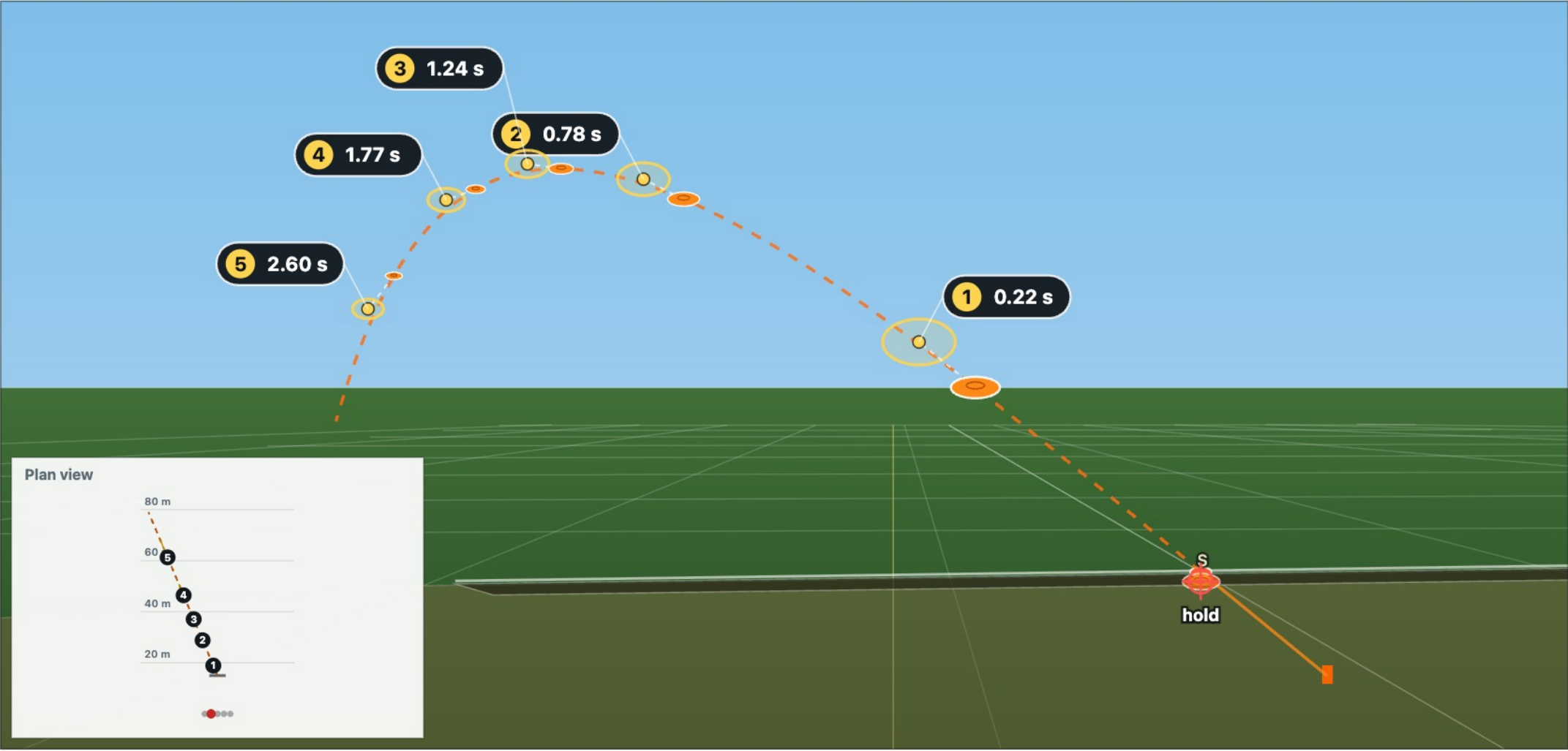
1 Release 0.22 s after release 58 cm ≈ 5.2 clays 37 cm left · 44 cm up -0.98° lateral · +1.15° vertical clay at 22 m · height 1.2 m · real separation 2.6 m first glimpse · within reach	2 Early 0.56 s after release 50 cm ≈ 4.5 clays 41 cm left · 28 cm up -0.75° lateral · +0.52° vertical clay at 31 m · height 2.4 m · real separation 4.0 m quick shot · within reach	3 Good window 0.85 s after release 45 cm ≈ 4.1 clays 44 cm left · 6 cm up -0.65° lateral · +0.09° vertical clay at 39 m · height 3.1 m · real separation 5.5 m the comfortable shot · at the limit	4 At the limit 1.17 s after release 58 cm ≈ 5.3 clays 49 cm left · 31 cm down -0.58° lateral · -0.37° vertical clay at 48 m · height 3.4 m · real separation 7.4 m last useful shot · at the limit	5 You are late 1.78 s after release 158 cm ≈ 14.4 clays 57 cm left · 148 cm down -0.50° lateral · -1.30° vertical clay at 65 m · height 2.9 m · real separation 11.7 m no longer reaching well · beyond useful range
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Station 2 · Machine 1

angle -23° · height 3.5 m at 10 m · fall 70 m · 75 km/h · elevation 21.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -13° horizontally and +6° vertically



Plan view



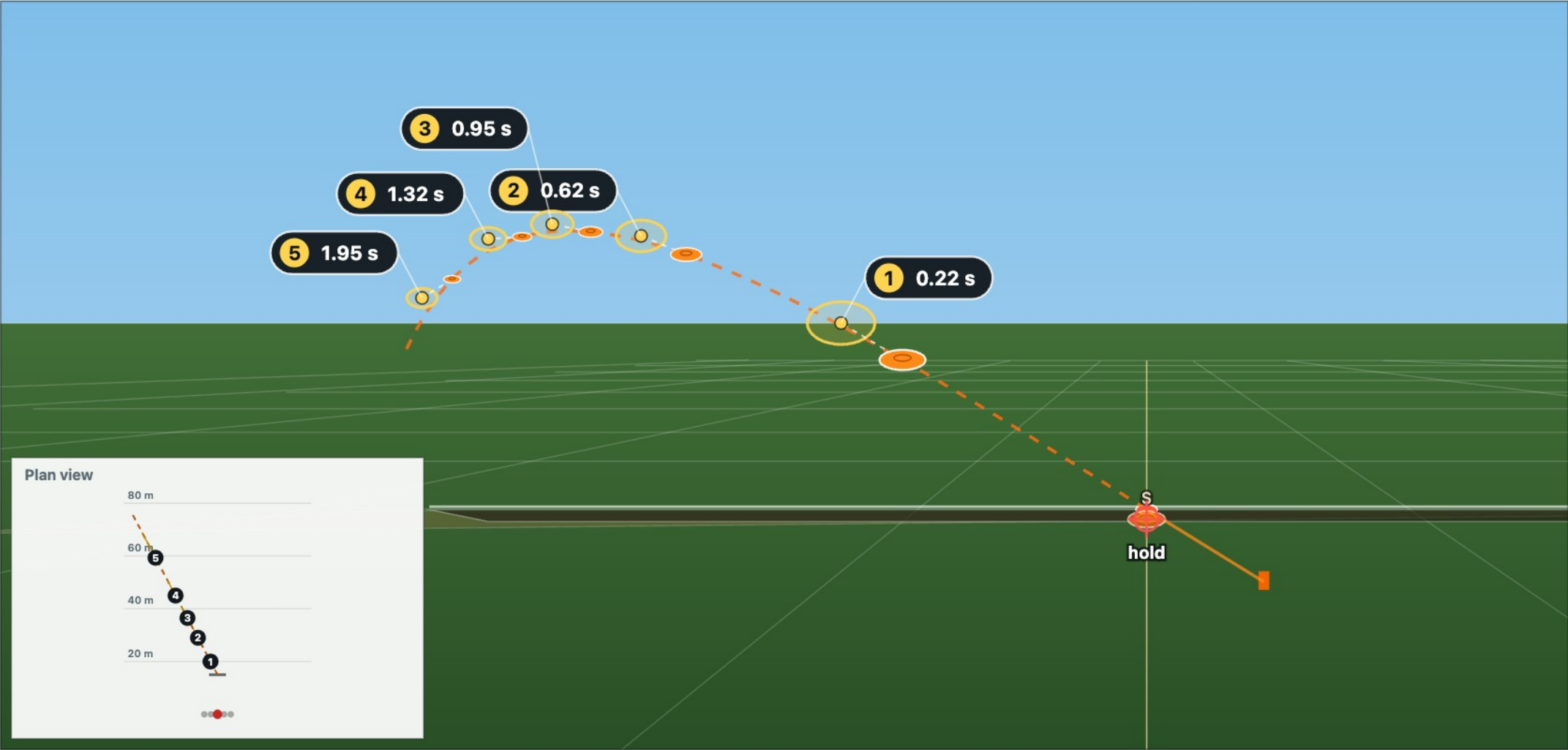
1 Release 0.22 s after release 79 cm ≈ 7.1 clays 62 cm left · 48 cm up -1.88° lateral · +1.45° vertical clay at 19 m · height 1.7 m · real separation 1.5 m first glimpse · within reach	2 Early 0.78 s after release 74 cm ≈ 6.7 clays 69 cm left · 26 cm up -1.35° lateral · +0.52° vertical clay at 29 m · height 4.7 m · real separation 2.5 m quick shot · within reach	3 Good window 1.24 s after release 75 cm ≈ 6.8 clays 75 cm left · 5 cm down -1.13° lateral · -0.08° vertical clay at 38 m · height 6.2 m · real separation 3.5 m the comfortable shot · within reach	4 At the limit 1.77 s after release 102 cm ≈ 9.3 clays 82 cm left · 60 cm down -0.98° lateral · -0.72° vertical clay at 48 m · height 6.9 m · real separation 4.8 m last useful shot · at the limit	5 You are late 2.60 s after release 227 cm ≈ 20.6 clays 97 cm left · 205 cm down -0.87° lateral · -1.84° vertical clay at 64 m · height 5.6 m · real separation 7.7 m no longer reaching well · beyond useful range
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Station 3 · Machine 1

angle -28° · height 2.0 m at 10 m · fall 70 m · 95 km/h · elevation 12.3°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -11° horizontally and +5° vertically



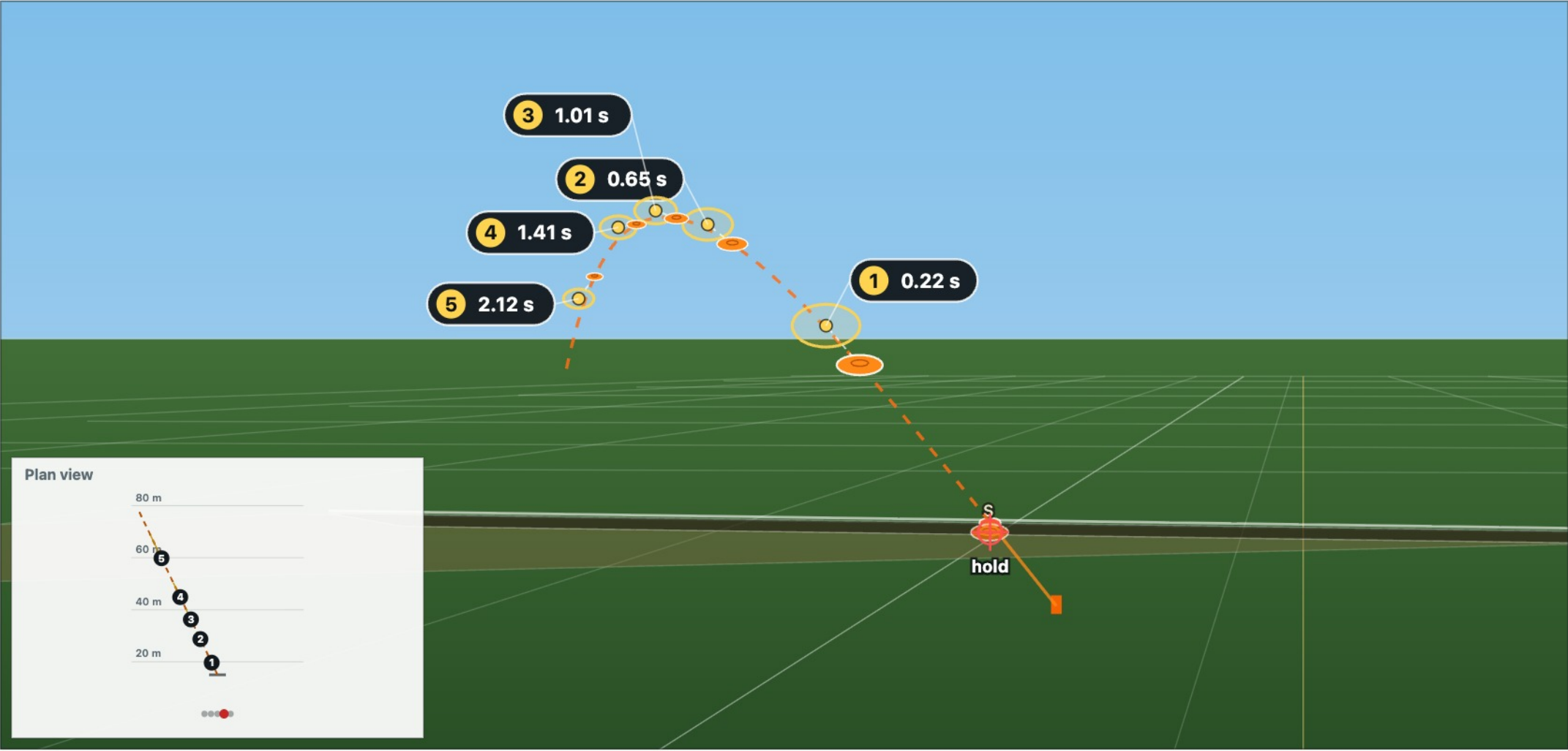
1 Release 0.22 s after release 86 cm ≈ 7.8 clays 75 cm left · 42 cm up -2.12° lateral · +1.19° vertical clay at 20 m · height 1.3 m · real separation 2.2 m first glimpse · within reach	2 Early 0.62 s after release 86 cm ≈ 7.8 clays 82 cm left · 25 cm up -1.57° lateral · +0.48° vertical clay at 30 m · height 2.8 m · real separation 3.5 m quick shot · within reach	3 Good window 0.95 s after release 90 cm ≈ 8.1 clays 90 cm left · 1 cm up -1.34° lateral · +0.01° vertical clay at 38 m · height 3.6 m · real separation 4.9 m the comfortable shot · at the limit	4 At the limit 1.32 s after release 107 cm ≈ 9.8 clays 99 cm left · 41 cm down -1.19° lateral · -0.50° vertical clay at 48 m · height 4.0 m · real separation 6.7 m last useful shot · at the limit	5 You are late 1.95 s after release 203 cm ≈ 18.5 clays 119 cm left · 165 cm down -1.07° lateral · -1.48° vertical clay at 64 m · height 3.2 m · real separation 10.9 m no longer reaching well · beyond useful range
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Station 4 · Machine 1

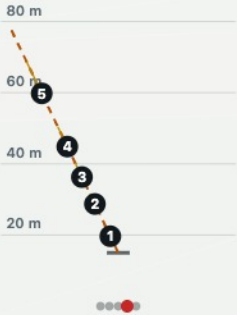
angle -26° · height 2.4 m at 10 m · fall 70 m · 87 km/h · elevation 14.6°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -6° horizontally and +5° vertically



Plan view



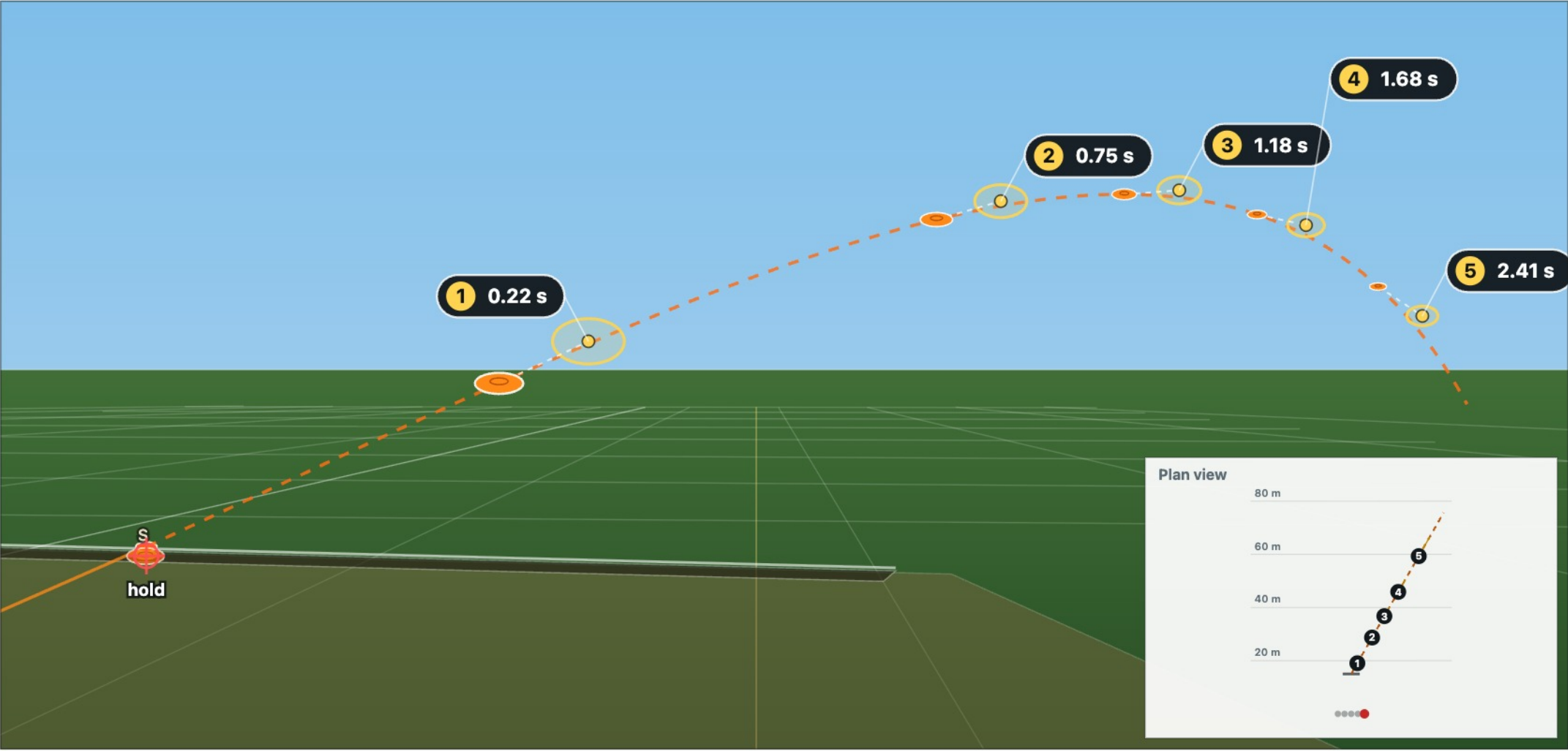
1 Release 0.22 s after release 61 cm ≈ 5.5 clays 41 cm left · 45 cm up -1.16° lateral · +1.27° vertical clay at 20 m · height 1.4 m · real separation 2.0 m first glimpse · within reach	2 Early 0.65 s after release 52 cm ≈ 4.8 clays 45 cm left · 27 cm up -0.85° lateral · +0.51° vertical clay at 30 m · height 3.3 m · real separation 3.2 m quick shot · within reach	3 Good window 1.01 s after release 49 cm ≈ 4.4 clays 49 cm left · 1 cm up -0.73° lateral · +0.02° vertical clay at 39 m · height 4.2 m · real separation 4.4 m the comfortable shot · at the limit	4 At the limit 1.41 s after release 69 cm ≈ 6.3 clays 54 cm left · 43 cm down -0.64° lateral · -0.52° vertical clay at 48 m · height 4.7 m · real separation 6.1 m last useful shot · at the limit	5 You are late 2.12 s after release 189 cm ≈ 17.2 clays 65 cm left · 178 cm down -0.57° lateral · -1.58° vertical clay at 64 m · height 3.9 m · real separation 9.9 m no longer reaching well · beyond useful range
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Station 5 · Machine 1

angle +30° · height 2.9 m at 10 m · fall 70 m · 80 km/h · elevation 17.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +19° horizontally and +6° vertically



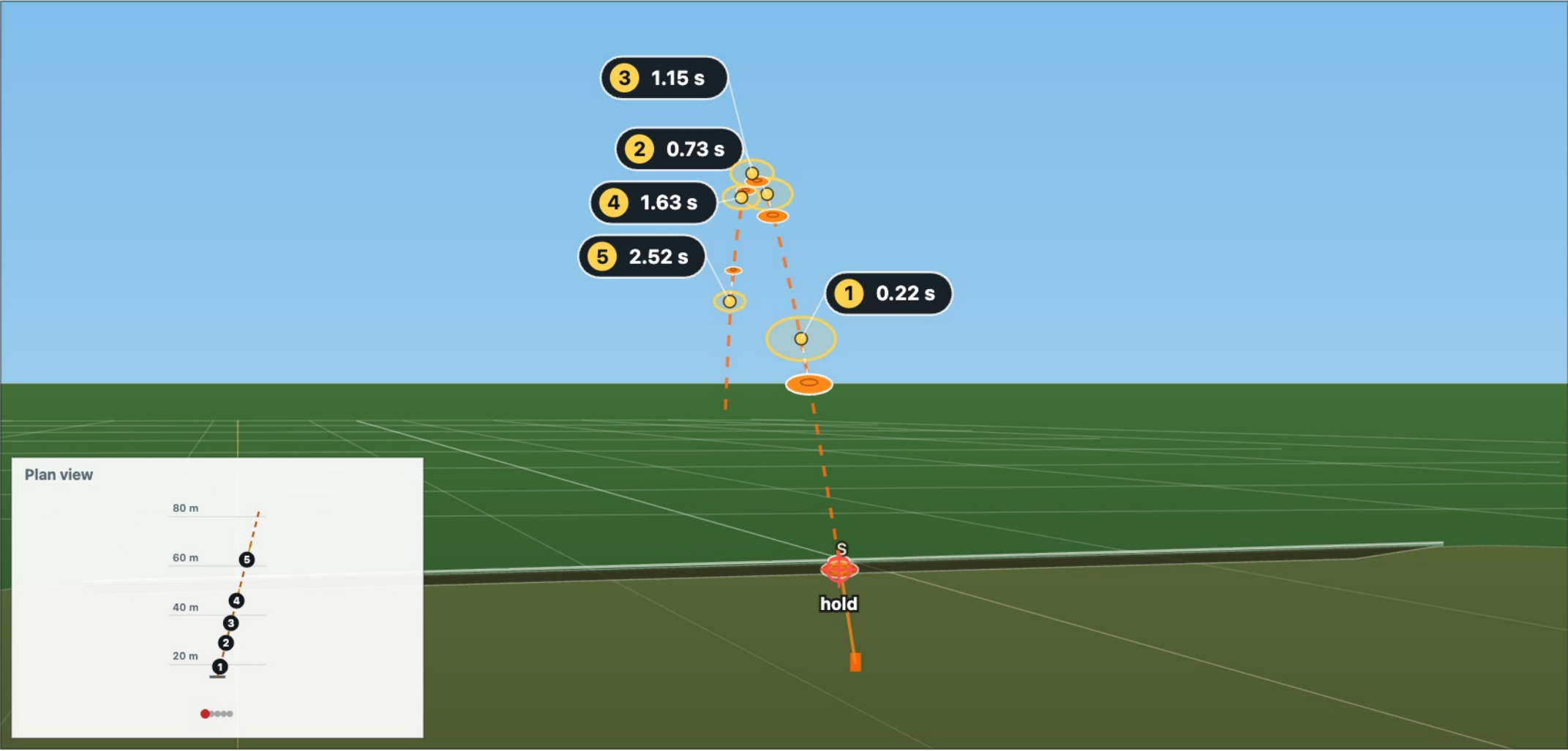
1 Release 0.22 s after release 109 cm ≈ 9.9 clays 99 cm right · 45 cm up +2.95° lateral · +1.33° vertical clay at 19 m · height 1.6 m · real separation 1.7 m first glimpse · within reach	2 Early 0.75 s after release 112 cm ≈ 10.2 clays 110 cm right · 24 cm up +2.17° lateral · +0.47° vertical clay at 29 m · height 4.1 m · real separation 2.7 m quick shot · within reach	3 Good window 1.18 s after release 120 cm ≈ 10.9 clays 119 cm right · 7 cm down +1.82° lateral · -0.11° vertical clay at 38 m · height 5.3 m · real separation 3.8 m the comfortable shot · within reach	4 At the limit 1.68 s after release 145 cm ≈ 13.2 clays 132 cm right · 60 cm down +1.58° lateral · -0.72° vertical clay at 48 m · height 5.8 m · real separation 5.3 m last useful shot · at the limit	5 You are late 2.41 s after release 245 cm ≈ 22.3 clays 155 cm right · 190 cm down +1.41° lateral · -1.73° vertical clay at 63 m · height 4.6 m · real separation 8.4 m no longer reaching well · beyond useful range
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Station 1 · Machine 1

angle +14° · height 3.4 m at 10 m · fall 70 m · 75 km/h · elevation 20.8°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned -2° horizontally and +6° vertically



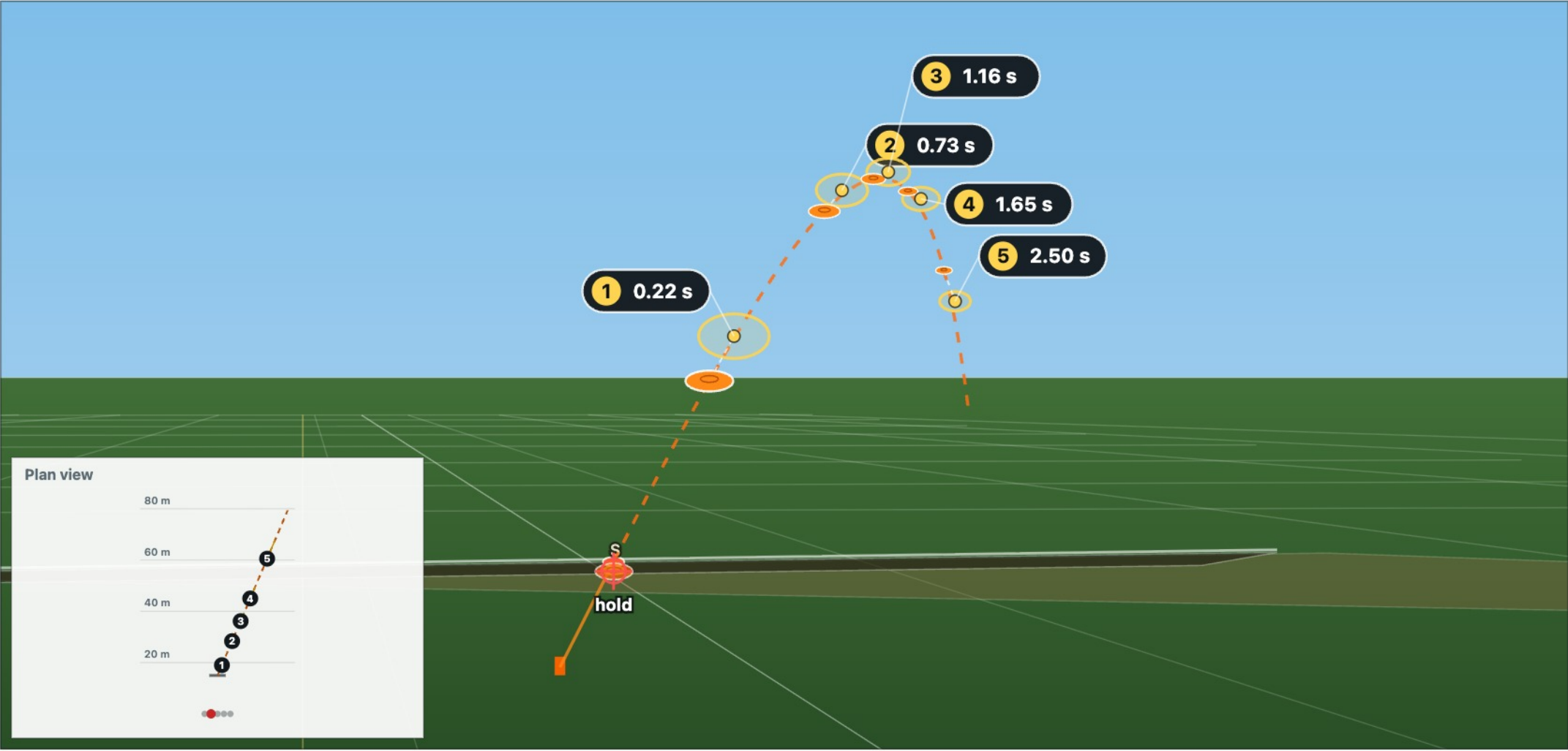
1 Release 0.22 s after release 52 cm ≈ 4.8 clays 10 cm left · 51 cm up -0.27° lateral · +1.47° vertical clay at 20 m · height 1.7 m · real separation 1.7 m first glimpse · within reach	2 Early 0.73 s after release 33 cm ≈ 3.0 clays 10 cm left · 31 cm up -0.20° lateral · +0.59° vertical clay at 30 m · height 4.5 m · real separation 2.7 m quick shot · within reach	3 Good window 1.15 s after release 11 cm ≈ 1.0 clays 11 cm left · 2 cm up -0.17° lateral · +0.03° vertical clay at 38 m · height 5.9 m · real separation 3.6 m the comfortable shot · at the limit	4 At the limit 1.63 s after release 49 cm ≈ 4.5 clays 12 cm left · 48 cm down -0.15° lateral · -0.57° vertical clay at 48 m · height 6.7 m · real separation 4.9 m last useful shot · at the limit	5 You are late 2.52 s after release 203 cm ≈ 18.5 clays 15 cm left · 203 cm down -0.13° lateral · -1.79° vertical clay at 65 m · height 5.7 m · real separation 8.0 m no longer reaching well · beyond useful range
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Station 2 · Machine 1

angle +23° · height 3.3 m at 10 m · fall 70 m · 76 km/h · elevation 20.2°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +5° horizontally and +6° vertically



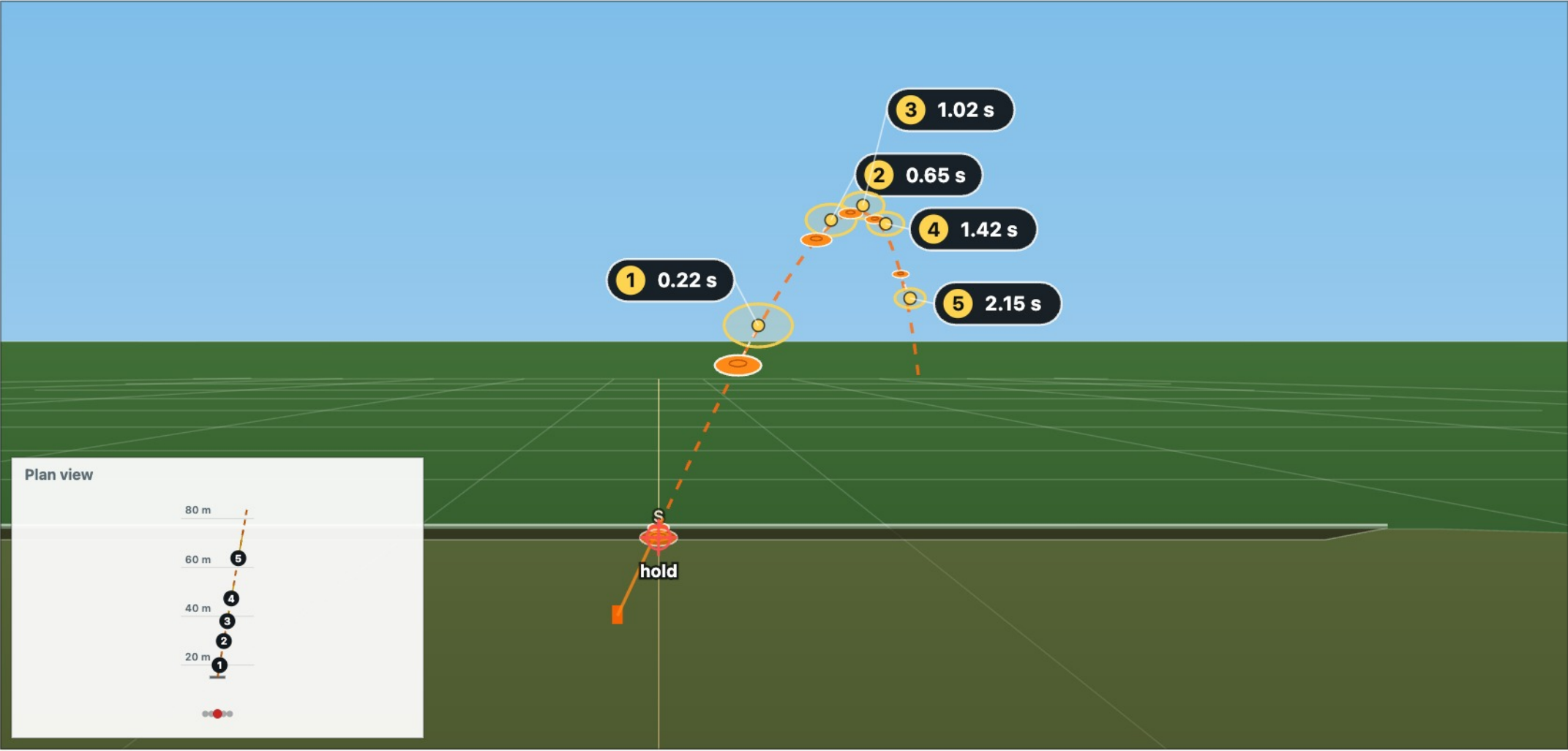
1 Release 0.22 s after release 56 cm ≈ 5.1 clays 28 cm right · 49 cm up +0.83° lateral · +1.44° vertical clay at 19 m · height 1.7 m · real separation 1.6 m first glimpse · within reach	2 Early 0.73 s after release 42 cm ≈ 3.8 clays 31 cm right · 29 cm up +0.60° lateral · +0.55° vertical clay at 30 m · height 4.4 m · real separation 2.6 m quick shot · within reach	3 Good window 1.16 s after release 34 cm ≈ 3.1 clays 34 cm right · 1 cm down +0.51° lateral · -0.01° vertical clay at 38 m · height 5.8 m · real separation 3.6 m the comfortable shot · at the limit	4 At the limit 1.65 s after release 63 cm ≈ 5.7 clays 37 cm right · 51 cm down +0.44° lateral · -0.61° vertical clay at 48 m · height 6.6 m · real separation 4.9 m last useful shot · at the limit	5 You are late 2.50 s after release 205 cm ≈ 18.7 clays 44 cm right · 201 cm down +0.39° lateral · -1.78° vertical clay at 64 m · height 5.5 m · real separation 8.1 m no longer reaching well · beyond useful range
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Station 3 · Machine 1

angle +10° · height 2.5 m at 10 m · fall 70 m · 86 km/h · elevation 15.2°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +4° horizontally and +5° vertically



Plan view



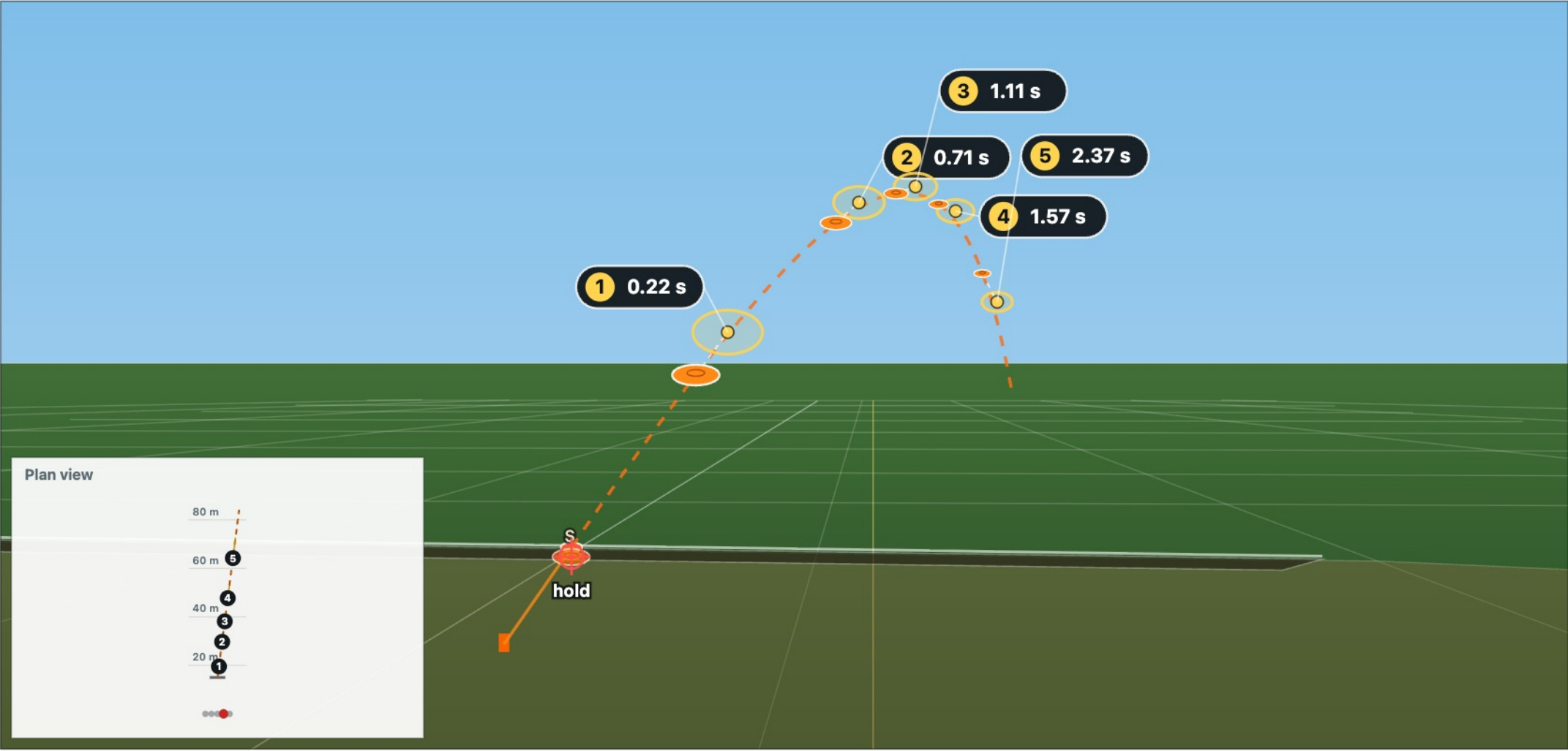
1 Release 0.22 s after release 51 cm ≈ 4.6 clays 24 cm right · 45 cm up +0.70° lateral · +1.29° vertical clay at 20 m · height 1.4 m · real separation 1.9 m first glimpse · within reach	2 Early 0.65 s after release 38 cm ≈ 3.5 clays 27 cm right · 27 cm up +0.51° lateral · +0.52° vertical clay at 30 m · height 3.4 m · real separation 3.1 m quick shot · within reach	3 Good window 1.02 s after release 29 cm ≈ 2.6 clays 29 cm right · 1 cm up +0.43° lateral · +0.02° vertical clay at 38 m · height 4.4 m · real separation 4.3 m the comfortable shot · at the limit	4 At the limit 1.42 s after release 54 cm ≈ 4.9 clays 32 cm right · 43 cm down +0.38° lateral · -0.52° vertical clay at 48 m · height 4.9 m · real separation 5.9 m last useful shot · at the limit	5 You are late 2.15 s after release 184 cm ≈ 16.7 clays 38 cm right · 179 cm down +0.34° lateral · -1.60° vertical clay at 64 m · height 4.1 m · real separation 9.7 m no longer reaching well · beyond useful range
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Station 4 · Machine 1

angle +7° · height 3.0 m at 10 m · fall 70 m · 79 km/h · elevation 18.2°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +7° horizontally and +6° vertically



Plan view



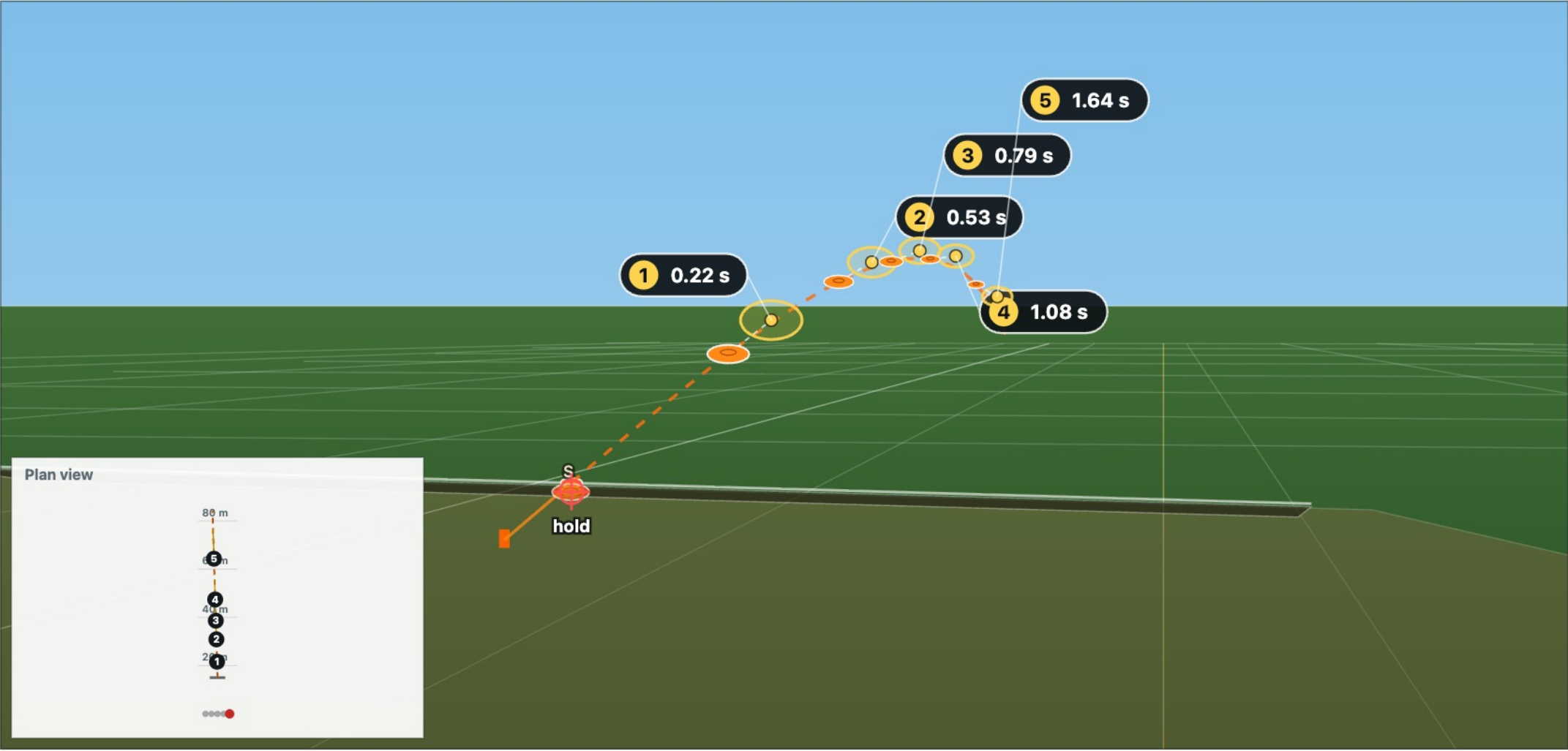
1 Release 0.22 s after release 60 cm ≈ 5.5 clays 37 cm right · 47 cm up +1.08° lateral · +1.37° vertical clay at 20 m · height 1.6 m · real separation 1.7 m first glimpse · within reach	2 Early 0.71 s after release 49 cm ≈ 4.5 clays 41 cm right · 28 cm up +0.79° lateral · +0.53° vertical clay at 30 m · height 4.0 m · real separation 2.8 m quick shot · within reach	3 Good window 1.11 s after release 44 cm ≈ 4.0 clays 44 cm right · 1 cm down +0.67° lateral · −0.01° vertical clay at 38 m · height 5.3 m · real separation 3.9 m the comfortable shot · at the limit	4 At the limit 1.57 s after release 70 cm ≈ 6.3 clays 49 cm right · 50 cm down +0.58° lateral · −0.59° vertical clay at 48 m · height 5.9 m · real separation 5.3 m last useful shot · at the limit	5 You are late 2.37 s after release 202 cm ≈ 18.3 clays 58 cm right · 193 cm down +0.52° lateral · −1.72° vertical clay at 64 m · height 4.9 m · real separation 8.6 m no longer reaching well · beyond useful range
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Station 5 · Machine 1

angle -2° · height 1.5 m at 10 m · fall 70 m · 109 km/h · elevation 9.1°

THE LEADS ON THIS SHEET ARE VALID ONLY WITH THESE SETTINGS

cartridge 400 m/s · rib 0% (0.00°) · wind 0.0 m/s · delay 0.03 s · pattern 1 3/4 · hold neutral
framing 2.16x over the base field of 48° vertical · view turned +7° horizontally and +4° vertically



1 Release 0.22 s after release 73 cm ≈ 6.6 clays 59 cm right · 43 cm up +1.52° lateral · +1.10° vertical clay at 22 m · height 1.1 m · real separation 2.9 m first glimpse · within reach	2 Early 0.53 s after release 71 cm ≈ 6.4 clays 65 cm right · 28 cm up +1.18° lateral · +0.52° vertical clay at 31 m · height 2.1 m · real separation 4.5 m quick shot · within reach	3 Good window 0.79 s after release 71 cm ≈ 6.4 clays 70 cm right · 8 cm up +1.03° lateral · +0.12° vertical clay at 39 m · height 2.7 m · real separation 6.1 m the comfortable shot · at the limit	4 At the limit 1.08 s after release 82 cm ≈ 7.5 clays 78 cm right · 27 cm down +0.93° lateral · -0.32° vertical clay at 48 m · height 2.9 m · real separation 8.2 m last useful shot · at the limit	5 You are late 1.64 s after release 161 cm ≈ 14.6 clays 89 cm right · 134 cm down +0.79° lateral · -1.19° vertical clay at 65 m · height 2.5 m · real separation 12.7 m no longer reaching well · beyond useful range
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Read it before applying anything from this notebook on a shooting ground

This is an estimate, not a measurement

The trajectories, the times and the leads come from a simplified model that solves a visual parabola from the height and the range of the clay, with reduced gravity and a common visual release. It does not measure a real trap, it does not reproduce the full aerodynamics of the disc, and it does not replace a ballistic table for the cartridge you use.

The scheme tables reproduce values published by the federations, but this document does not certify any installation or any trajectory. In competition the authorised officials always decide.

The responsibility for the shot is yours

Use these figures as a starting point for observing and comparing, never as an instruction to follow blindly. Check your own numbers on the ground, with your gun, your cartridge and the conditions of the day, and correct whatever needs correcting.

This material does not replace in-person training, the rules of the ground, the help of an instructor or a gunsmith, or the legal requirements in force. Safety and shooting decisions are the responsibility of whoever pulls the trigger.

Before taking it to the ground

- Check that the scheme on the cover really is the one that is going to be thrown. Ask whoever is in charge instead of assuming it.
- The leads depend on the cartridge and the rib you set when generating the notebook. With different equipment, the figures change.
- Treat every number as a hypothesis you are going to test by shooting, not as the result you ought to get.

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Where everything is

Full book, simulator and source code:

www.robertocanales.com/proyectos#plato

rcanalescoder.github.io/Plato

rcanalesmora@gmail.com

The project is in beta

The book and the simulator are still under review and are shared so that anyone can try them, find errors and help improve them. If you find something that does not match your experience on the ground, the rules that apply to you, or the figures you measure yourself, write in: that kind of report is what makes the model better.